Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Arctic – Monitoring Forecast Centre
ARCTIC_ANALYSISFORECAST_PHY_002_001

Issue: 1.1

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	September 2023	All	Creation of the document	A. Melsom, L. Bertino	L. Bertino, ARC MFC leader
1.1	October 2025	All	New implementation for river runoff; new template	A. Melsom	L. Bertino

ACRONYM TABLE

Acronym	Description
CMEMS	Copernicus Marine Environment Monitoring Service
PUM	Product User Manual
MLD	Mixed Layer Depth
SST	Sea Surface Temperature
SLA	Sea Level Anomaly
T	Temperature
S	Salinity
SI, SIT	Sea Ice, Sea Ice Thickness
EAN	Estimated Accuracy Number
ARC MFC	ARCtic Monitoring and Forecasting Centre
RMSD	Root Mean Square Difference
MCC	Maximum Cross Correlation
HYCOM	The Hybrid Coordinate Ocean Model (HYCOM)
CICE	Community Ice CodE
ESMF	Earth System Modeling Framework
GloFAS	Global Flood Awareness System
ECMWF	European Centre for Medium-range Weather Forecasts
NERSC	Nansen Environmental and Remote Sensing Center
Met Norway	Norwegian Meteorological Institute
UTC	Coordinated Universal Time
SVP	Surface Velocity Program
AVHRR	Advanced Very-High-Resolution Radiometer
AWI	Alfred Wegener Institut
CPOM	UK Centre for Polar Observation and Modelling
DTU	Danmarks Tekniske Universitet (Technical University of Denmark)
SSMIS	Special Sensor Microwave Imager / Sounder
CTD	Conductivity, Temperature, and Depth
OSISAF	Ocean and Sea Ice Satellite Application Facility
CS2SMOS	CRyosat-2 / Soil Moisture and Ocean Salinity
OSTIA	Operational Sea Surface Temperature and Ice Analysis
NOAA	National Oceanic and Atmospheric Administration

KEY DEFINITIONS

Geostrophic current: The current resulting from the balance between the horizontal pressure gradient and the Coriolis force.

Low pass boxcar filter: A filter applying the boxcar function, which is zero over the entire real line except for a single interval where it is equal to a constant > 0.

Mixed layer depth: The depth at which the density has increased by a magnitude of 0.03 kg/m^3 from its value at a depth of 10m. This represents the layer which is close to being homogeneous due to mixing by surface fluxes, notably the wind stress.

z-isopycnic surfaces: A hybrid set of surfaces based on distance to the surface at rest (z) and surfaces where the density is constant (isopycnic).

Isopotential surfaces: Surfaces along which there is no component of the gravitational force, i.e., surfaces with the same geopotential. Note that this is (nearly) identical to standard z surfaces.

RELEVANT DOCUMENT LIST

CMEMS ARC MFC PUM	Product User Manual for Arctic Ocean Physical and BGC Analysis and Forecasting Products[1]
The TOPAZ system	TOPAZ4: an ocean-sea ice data assimilation system for the North Atlantic and Arctic. doi:10.5194/os-8-633-2012[2]

[1] https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-ALL.pdf
[2] https://os.copernicus.org/articles/8/633/2012/

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document covers the following product:

- **ARCTIC_ANALYSISFORECAST_PHY_002_001**, which is the main product from the Copernicus Marine Environment Monitoring Service (CMEMS) for real-time forecasts of ocean circulation and sea ice in the Arctic region.

The model has a native irregular horizontal grid, and vertical layers with a time-dependent thicknesses. Hence, in order to provide a product that is accessible to the general user, results are interpolated onto a 6.25 km horizontal grid using a polar stereographic map projection, and onto 40 vertical z-levels.

The ARCTIC_ANALYSISFORECAST_PHY_002_001 product delivers four datasets:

- cmems_mod_arc_phy_anfc_6km_detided_PT1H-I (hourly instantaneous values)
- cmems_mod_arc_phy_anfc_6km_detided_PT6H-m (6-hourly mean values)
- cmems_mod_arc_phy_anfc_6km_detided_P1D-m (daily mean values)
- cmems_mod_arc_phy_anfc_6km_detided_P1M-m (monthly mean values)

The model domain is displayed in Figure 1 (see Section III). The 6-hourly, daily and monthly datasets give results in three dimensions for temperature, salinity and velocity vector components. Results are also provided for mixed layer depth, sea surface height, and concentration, thickness, age and drift speed for sea ice, the thickness of snow on the sea ice, and the surface albedo. Results from the hourly dataset are limited to surface values for these variables.

For the present qualification results, the cmems_mod_arc_phy_anfc_6km_detided_P1D-m dataset was investigated. Quality metrics, including estimated accuracy numbers, were computed on the 6.25 km, 40 level output grid, as available to the CMEMS users. Unfortunately, there are no regular provisions of observations at sub-daily frequencies that are relevant for qualifying our hourly and 6-hourly datasets. While we have not specifically included accuracy numbers for the monthly product, its quality can be deduced from the depictions that cover one year for the daily mean results. Where relevant, we also discuss seasonal changes in the quality. More technical details on this product are available in the Product User Manual (PUM).

I.2 Summary of the results

Validation of products from the operational forecasts has been performed using remote sensing datasets for sea ice variables (see Table 1a in Section I.3), and a combination of in situ measurements and remote sensing data for physical ocean variables (see Table 1b, c in Section I.3).

Based on the detailed results provided in Section IV, this Arctic MFC product is in good general agreement with observations. The following list provides more detail, including issues for which there remains room for improvement:

- The sea ice concentration has a small positive bias, which is mainly attributed to sea ice concentrations being overestimated when observations are in the range [0.4, 0.9]
- The sea ice edge is slightly overestimated by the model by about 100 km on average, suggesting that the shape of the edge is more undulating in the model.

- The model results for sea ice thickness have a too narrow distribution, as categories for smallest and largest thicknesses are underrepresented.
- In situ and remote sensing data reveal that model result for sea surface temperature are of a high quality.
- Displacement distances due to sea ice drift are very similar in model and observations, but differences in drift direction are notable.
- The near-surface drift in the model is too weak, and the observed drift direction is also misrepresented by the model results.
- The averaged results for mixed layer depth in model and observations agree well, particularly in the Nordic Seas. However, when stratification is weak in winter differences can be large.
- The observed vertical distribution of heat and salt are well captured in the model results.

I.3 Estimated Accuracy Numbers

The results shown below are based on analyses of model daily average results from a one-year period 2022-07-01 – 2023-06-30, based on weekly analyses with Ensemble Kalman filter assimilation. Bias (mean difference/mean error) is calculated by subtracting observations from model results (i.e., model value – observed value). The bias results are supplemented by values for the Root Mean Square Differences (RMSD) between model results and observations.

Table 1a: *Estimated accuracy numbers for sea ice variables. RMSD represents daily mean RMSDs. Units are specified in the 'Variable' column. Sub-domain regions are displayed in Figure 1(a). The sources of observational data are detailed in Table 5. (*) Results for sea ice thickness are not available during summer. In contrast to the other variables, sea ice edge length has only one value for each region, and for each day (see section IV 2.1). The bias simply becomes the difference between the modelled and observed edge length, and there is no RMSD.*

Period: July 2022 – June 2023				
Variable	**Region**	**Supporting observations**	**Best estimate**	
			RMSD	**Bias (model-observation)**
Sea ice concentration (area fraction)	**Full domain**	**SSMIS**	**0.13**	**0.04**
	American s.		0.12	0.04
	European s.		0.13	0.04
	Asian s.		0.13	0.04
Sea ice edge length (Kkm)	**Full domain**		**-**	**0.11**
	American s.		-	-0.01
	European s.		-	0.13
	Asian s.		-	-0.01
Sea ice edge position (km)	**Full domain**		**78**	**1**
	American s.		88	-3
	European s.		56	2
	Asian s.		65	10
Sea ice thickness(*) (m)	**Full domain**	**SMOS**	**0.33**	0.11
		Cryosat-SIRAL	**0.78**	-0.31
Sea ice drift displacement (km)	**Full domain**	**SAR**	**5.4**	**-0.2**
	American s.		4.2	0.7
	European s.		6.7	-1.1
	Asian s.		5.7	-1.3

Table 1b: *Estimated accuracy numbers for 2D ocean variables. Sub-domain regions are displayed in Figure 1(b).*

Period: July 2022 – June 2023				
Variable	**Region**	**Supporting observations**	**Best estimate**	
			RMSD	**Bias (model-observation)**
Near-surface drift distance (km)	**Full domain**	**SVP drifters**	**18**	**-8**
	Nordic Seas		17	-9
	Barents Sea		14	-8
Surface geostrophic velocity component (cm/s)	**Full domain**	**Sentinel-6a**	**10**	**-6**
		Jason-3	**9**	**-5**
		Cryosat-2	**10**	**-5**
	Nordic Seas	Sentinel-6a	8	-5
		Jason-3	9	-5
		Cryosat-2	10	-7
Sea surface temperature (K)	**Full domain**	**SVP drifters**	**0.7**	**0.1**
		AVHRR	**0.8**	**0.1**
	Nordic Seas	SVP drifters	0.9	0.1
		AVHRR	0.7	0.1
	Barents Sea	SVP drifters	0.6	0.0
		AVHRR	0.7	0.3
Mixed layer depth (m)	**Full domain**	**Profiling instruments**	**52**	20
	Nordic Seas		36	3
Sea level anomaly (cm)	**Full domain**	**Sentinel-6a**	**7**	**-2**
		Jason-3	**7**	**-2**
		Cryosat-2	**8**	**-2**
	Nordic Seas	Sentinel-6a	4	1
		Jason-3	4	0
		Cryosat-2	5	1

Table 1c: *Estimated accuracy numbers for 3D ocean variables. Sub-domain regions are displayed in Figure 1 (b).*

Period: July 2022 – June 2023					
Variable	**Region**	**Supporting observations**	**Layer (m)**	**Best estimate**	
				RMSD	**Bias (model-observation)**
Potential temperature (K)	**Full domain**	**Profiling instruments**	**0-5**	**0.35**	**-0.03**
			5-100	**0.37**	**-0.10**
			100-300	**0.41**	**0.11**
			300-800	**0.56**	**0.17**
			800-2000	**0.36**	**-0.15**
	Nordic Seas		0-5	0.33	-0.04
			5-100	0.39	-0.15
			100-300	0.42	0.06
			300-800	0.43	0.01
			800-2000	0.27	-0.22
Salinity (‰)	**Full domain**		**0-5**	**0.11**	**-0.01**
			5-100	**0.09**	**-0.01**
			100-300	**0.05**	**-0.00**
			300-800	**0.03**	**0.00**
			800-2000	**0.02**	**0.00**
	Nordic Seas		0-5	0.07	-0.00
			5-100	0.05	-0.01
			100-300	0.03	-0.00
			300-800	0.02	0.01
			800-2000	0.01	0.00

II PRODUCTION SYSTEM DESCRIPTION

Production centre name

Norwegian Meteorological Institute

Production system name

Arctic near real time forecast and analysis (Copernicus Marine Service name: ARCTIC_ANALYSISFORECAST_PHY_002_001)

Production centre description for the version covered by this document

The product is produced by the TOPAZ5 data assimilation system for ocean circulation and sea ice. TOPAZ5 is a coupled ocean and sea ice configuration which applies an advanced sequential data assimilation method. Details are given in Table 2 below.

Table 2: *Overview of TOPAZ5*

System component	Model name	Version
Ocean circulation model	The Hybrid Coordinate Ocean Model (HYCOM)	v.2.2.98
Sea ice model	The CICE sea ice model	v.5.1
Assimilation software	Deterministic Ensemble Kalman Filter (DEnKF)	v.2.11
Coupler	Earth System Modelling Framework (ESMF)	v.8.0.1

TOPAZ5 is configured with a z-isopycnal grid in the vertical dimension, and a staggered and slightly stretched grid in the horizontal dimensions. Prior to dissemination, the results are interpolated onto a 6.25 km resolution grid on a polar stereographic map projection. Moreover, results are given for 40 constant depth z-levels.

In this document the analyses are performed by comparing results from the daily mean product to observations. The accompanying reanalysis product has been validated by Xie et al. (2017).

For the TOPAZ5 simulations, the following model forcings are applied

- Atmospheric forcing: Data from ECMWF's HRES (High Resolution) Model. The data are on a global regular lat/lon grid (0.1° x 0.1°), with a temporal resolution of 6 h, interpolated to the TOPAZ5 mesh.
- Lateral forcing: Open boundary values are received from the Copernicus Marine Service' global forecast and analysis model for ocean circulation, GLOBAL_ANALYSISFORECAST_PHY_001_024.
- River forcing: Fresh water fluxes are provided by daily forecasts (with up to a 30-day lead time) from the Global Flood Awareness System (GloFAS) v4.0, which is a global hydrological forecast

and monitoring system, jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts (ECMWF)[3].

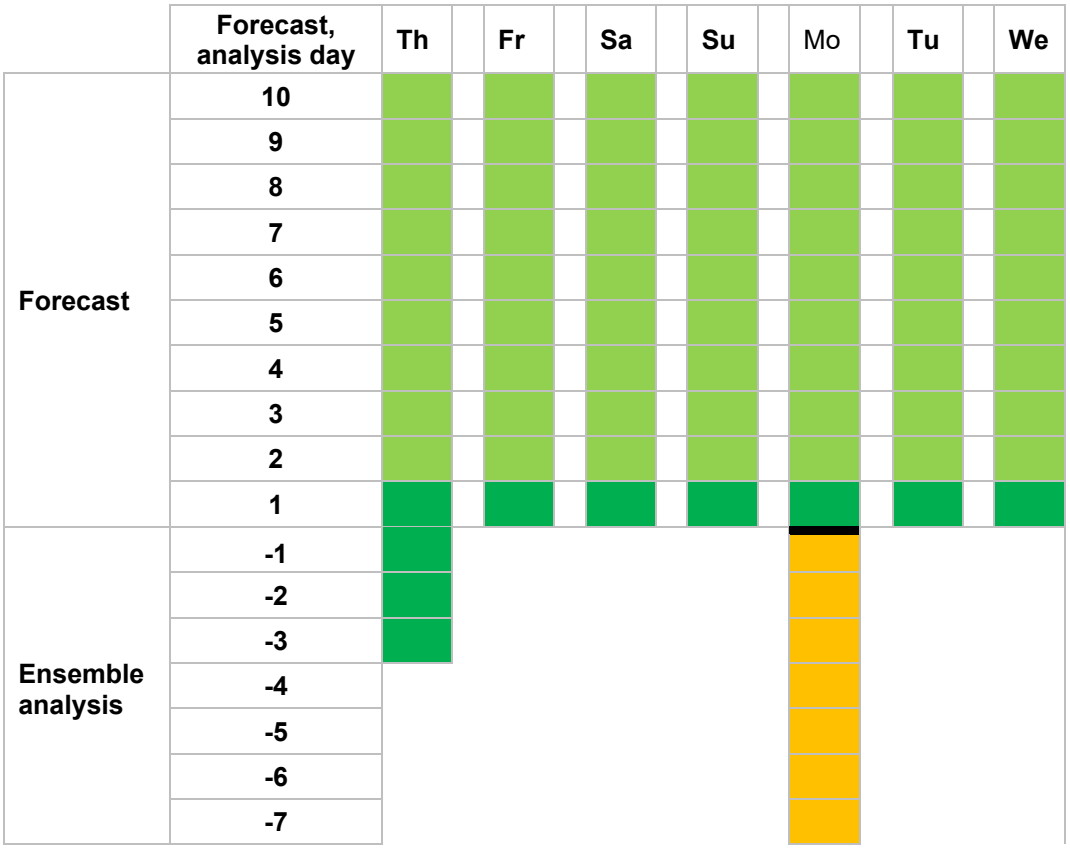
Observations that are applied during assimilation are described in Section II.2.

TOPAZ5 was developed and is maintained by the Nansen Environmental and Remote Sensing Center (NERSC[4]. It is run operationally for Copernicus Marine Service Arctic Monitoring and Forecasting production at the Norwegian Meteorological Institute (MET Norway[5]).

II.1 TOPAZ5 production cycle

TOPAZ5 is configured to run in a weekly cycle. The data assimilation step is run weekly on Thursdays, based on a 100-member ensemble that was produced the preceding Monday. Thus, the analysis has a valid time at 00 UTC on Monday. The Thursday forecasts are then run with updated atmospheric forcing

Table 3: *Illustration of the schedule for the TOPAZ5 system. The vertical axis corresponds to forecast lead time in days. Green indicates forecast simulation periods, while yellow corresponds to the time period for the 100-member Ensemble Kalman filter analysis. Dark green indicates forecasts that are kept as part of the "best estimate" product until they are superseded by the results from the Thursday simulation (labelled 'Th'), when a new analysis valid for 00 UTC on Monday initializes the simulation. See the text for further details.*



[3] For details see https://confluence.ecmwf.int/display/CEMS/Global+Flood+Awareness+System

[4] https://nersc.no/

[5] https://met.no/

fields from 00 UTC on Monday until 12 UTC on the simulation day (Thursday). The atmospheric 12 UTC forecast is then applied for the remaining 9.5 simulation days until 00 UTC on Sunday of the following week. For the remaining days of the week, new ocean and sea ice forecasts are initialized at 00 UTC from the preceding day's forecast, then forced with atmospheric results forecast at 00 UTC for the initial 12 hours, and updated atmospheric forecasts at 12 UTC for the final 9.5 simulation days. A 10-member ensemble is used for forecasting. The Copernicus Marine Service then disseminates these ensemble averages.

II.2 Observational Data

Observational data are assimilated into the TOPAZ system as part of the weekly analysis. All data are assimilated, but instead of performing a "background check", the observation error is increased for observations that are too far from the model forecast. This moderates the impact of their assimilation. The datasets that are assimilated are listed in Table 4.

Table 4: *List of sources from which data is assimilated in TOPAZ5.*

Variable	Source	Product/ dataset
Sea ice concentration	OSISAF Passive microwave data	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001/ METNO-GLO-SEAICE_CONC-NORTH-L4-NRT-OBS
Sea ice drift	OSISAF Passive microwave data	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001/ METNO-GLO-SEAICE_DRIFT-NORTH-L4-NRT-OBS
Sea ice thickness	CS2SMOS (merged data from Cryosat-2 and SMOS)	SEAICE_ARC_PHY_L4_NRT_011_014/ esa_obs-si_arc_phy-sit_nrt_l4_multi_P1D-m
Sea surface temperature	OSTIA from various infrared and microwave sensors	SST_GLO_PHY_L4_NRT_010_043/ cmems_obs-sst_glo_phy_nrt_l4_P1D-m
Temperature	Profile data from various instruments (notably ARGO floats)	INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/ cmems_obs-ins_glo_phybgcwav_mynrt_na_irr ; INSITU_ARC_PHYBGCWAV_DISCRETE_MYNRT_013_031/ cmems_obs-ins_arc_phybgcwav_mynrt_na_irr
Salinity		
Sea level anomaly	Altimeter data, from Altika; Cryosat-2; Sentinel-3a, 3b, 6a; Jason-3	SEALEVEL_GLO_PHY_L3_NRT_OBSERVATIONS_008_044/ dataset-duacs-nrt-global-al-phy-l3; dataset-duacs-nrt-global-c2n-phy-l3; dataset-duacs-nrt-global-s3a-phy-l3; dataset-duacs-nrt-global-s3b-phy-l3; cmems_obs-sl_glo_phy-ssh_nrt_s6a-hr-l3-duacs_PT1S; cmems_obs-sl_glo_phy-ssh_nrt_j3n-l3-duacs_PT1S

III VALIDATION FRAMEWORK

The model domain is displayed in Figure 1. We present two versions of this map to introduce the sub-regions for which the present quality assessment is undertaken.

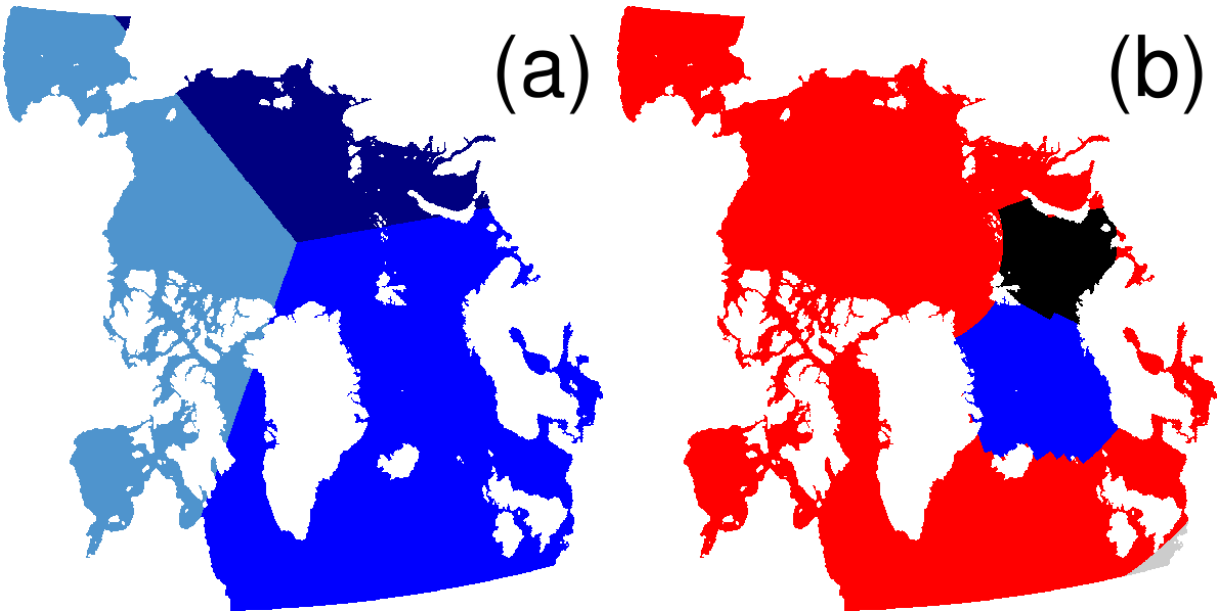


Figure 1: *Model domain. The model output is stored on a polar stereographic map projection with a resolution of 6.25km, covering the region that is displayed in this figure. The colour coding represents the division into sub-domains.* ***(a):*** *The American, European, and Asian sectors are shown in light, intermediate, and dark blue, respectively.* ***(b):*** *Nordic Seas in blue and Barents Sea in black.The domain is restricted to 50N in the south. The separation of the sectors in (a) are placed at 65°W (American/European), 55°E (European/Asian) and 175°E (Asian/American). The Nordic seas subdomain is inside the bounding box [25°W-20°E,61°N-79°N], while the Barents Sea subdomain is limited by [17°E-65°E, 68°N-80°N].*

The colours that identify the regions in Figure 1 are used in the corresponding line plots below, with red always referring to the full domain. Regions displayed in panel (a) are referenced in Figure 3-5,8, while panel (b) are referenced in Figure 10-11,2,15-20. The remaining figures (6-7.9,12,14,21) give results for the full domain only. Generally, sea ice data are available for nearly all parts of the model domain. In contrast, in situ data, and to some degree remote sensing data of the open ocean surface, is mostly limited to the Nordic Seas, with less observations available for the Barents Sea. These differences in data availability motivated the particular choice of regional sub-divisions in this document.

III.1 Model results

For the present quality assessment, model results were available as averages from 100-member ensemble simulations. Below is a list of the model variables and the sub-regions for which results are

provided:

American, European, and Asian sectors (Figure 1(a))

- Sea ice concentration
- Sea ice edge (position, length)
- Sea ice drift

Nordic Seas and Barents Sea (Figure 1(b))

- Ocean drift (near-surface drift; surface geostrophic current [Nordic Seas only])
- Sea surface temperature (drifting buoys; AVHRR remote sensing)
- Temperature in vertical layers [Nordic Seas only]
- Salinity in vertical layers [Nordic Seas only]
- Mixed layer depth [Nordic Seas only])
- Sea level anomalies

Quality metrics are also given for the full model domain. For the following variables, the results are presently only available for the full domain:

- Sea ice thickness (thickness of thin sea ice; thickness of thick sea ice)

Finally, due to lack of relevant observations, examination for the following variable is only given in the form of an expert assessment based on an inspection of the model results:

- Vertical velocity

Results from additional sub-domains may be included in later versions of this document. We will also consider supplying assessments for the remaining variables (sea ice albedo, sea ice snow thickness, sea ice age, bottom temperature) at a later time. Presently they are not included due to a lack of relevant observations.

III.2 External operational products

The sources of observational data that are considered for assessing the quality of the present product are listed in Table 5. The following considerations underpin our selection:

1. A preference for data that have an extensive coverage in space and time, so that the results from the analyses are as representative as possible of the quality throughout the domain, and for all seasons.
2. A preference for data that are independent from the data that are assimilated during production, to avoid artificially inflating the resulting assessment of quality.
3. A preference for separating the analyses by instrument, in order to make the quality assessment as homogeneous as possible.

Note that for most model variables, data sources that fully satisfy all these listed considerations are not available. Our choices were thus made using the points listed above as guidelines rather than strict exclusion criteria.

The processing that leads to quality assessment of model results for the sea ice thickness illustrates how the listed considerations are applied: The data that are used for computing the quality metrics are also assimilated by the model, but as a new hybrid product that is merged by post-processing the original data. Although this violates point 2 in the above list, there are no alternative observations for sea ice

thickness that can match the spatial and temporal coverage (point 1). But instead of comparing with the hybrid product that is assimilated, the quality analyses are split for the two instruments merged in the hybrid product, in compliance with point 3.

Finally, we choose to separate results from the three altimeters that are used in the quality assessment of sea level anomalies and geostrophic currents. Although these observations have a high degree of instrument homogeneity, the corresponding satellites have substantial differences in their orbits. Hence, the separation is made in accordance with point 3 above.

Table 5: *List of data sources applied in the validation described in in Section IV.*

Variable	Source	Product/ dataset
Sea ice concentration	OSISAF passive microwave data	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001/ METNO-GLO-SEAICE_CONC-NORTH-L4-NRT-OBS
Sea ice edge		
Sea ice thickness	AWI SMOS sea ice product (thin sea ice)	ftp.awi.de: sea_ice/product/smos/v3.3/nh
	CPOM data portal Cryosat-2 SIRAL data (thick sea ice)	http://www.cpom.ucl.ac.uk/csopr/seaice.php
Sea ice drift	DTU sea ice drift from SAR data	SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_006/ DTU-GLO-SEAICE_DRIFT-NORTH-L4-NRT-OBS
Sea surface temperature	IFREMER AVHRR Metop-B infrared data	eftp.ifremer.fr: /cersat-rt/project/osi-saf/data/sst/l3c/global/ glob_avhrr_metop_b
Near-surface drift	IFREMER distribution of SVP drifter data	ftp.ifremer.fr: ifremer/cersat/projects/myocean/sst-tac/insitu/data/v2/
Geostrophic velocity component	Altimeter data, from Cryosat-2; Jason-3; Sentinel-6a	SEALEVEL_GLO_PHY_L3_NRT_OBSERVATIONS_008_044/ dataset-duacs-nrt-global-c2n-phy-l3; cmems_obs-sl_glo_phy-ssh_nrt_j3n-l3-duacs_PT1S; cmems_obs-sl_glo_phy-ssh_nrt_s6a-hr-l3-duacs_PT1S
Mixed layer depth	Profile data from various instruments (notably ARGO buoys)	INSITU_ARC_PHYBGCWAV_DISCRETE_MYNRT_013_031/ cmems_obs-ins_arc_phybgcwav_mynrt_na_irr
Temperature		
Salinity		
Sea level anomaly	Altimeter data, from Cryosat-2; Jason-3; Sentinel-6a	SEALEVEL_GLO_PHY_L3_NRT_OBSERVATIONS_008_044/ dataset-duacs-nrt-global-c2n-phy-l3; cmems_obs-sl_glo_phy-ssh_nrt_j3n-l3-duacs_PT1S; cmems_obs-sl_glo_phy-ssh_nrt_s6a-hr-l3-duacs_PT1S

III.3 Validation products

In this document, validation results are presented as quality metrics of bias and RMSD. The biases are computed by subtracting observed values from the relevant model results.

Tables 6, 8-10, 12, 14-21 show means of daily bias, their ranges, and/or 5 and 95 percentiles (i.e., the lowest and highest bias values that remain after the top 5% and the bottom 5% of the sorted bias values have been discarded).

In the same tables, the mean RMSD values are shown, supplemented by the 90 percentiles, i.e., the maximum daily RMSD after the top 10% of the daily values have been discarded.

Furthermore, we show temporal evolutions as line plots to assess the temporal variability of metrics, paying particular attention to seasonal changes. A seven-day low pass boxcar filter[6] is applied to the time series of daily values prior to plotting.

We also provide quality assessments for two vector quantities, sea ice drift and near-surface drift. Most validation metrics have been developed for scalar quantities, and are not necessarily applicable to vectors. Hence, when we present validation results as “biases” and “RMSD”, these properties are defined slightly differently than for scalar quantities (see Figure 2).

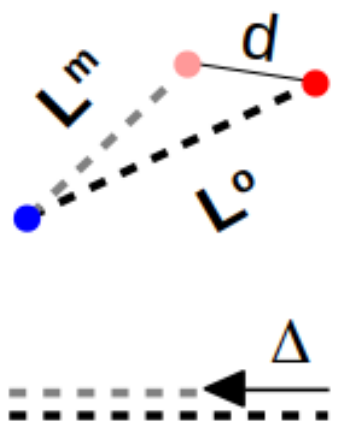


***Figure 2**: Illustration of drift quality metrics. The blue marker is the initial position from which drift is considered. Red and pink markers indicate the observed and modelled end position of drift for a given time period (here one day). The separation distance is the offset of the modelled end position from the observed end position, here labelled* d*. The drift length bias (Δ) is indicated by the arrow near the bottom. The bias is taken to be positive when the model displacement (L^m) is longer than the observed displacement (L^o). Thus, the bias is positive when the arrow points to the right (and negative in this illustration). Note that* d *can be interpreted as the model end position being somewhere on a circle centred on the red marker with radius* d*. This imposes a restriction on the angular difference between lines* L^o *and* L^m*.*

For this QUID version, model results were computed from a continuous time series, in the form of averages of a 100-member ensemble. We aim to supplement the present results with quality assessment for different forecast rangesin a future update for this product. Nevertheless, based on our experience with the predecessor of this product, quality is only weakly affected during 10-day forecasts, likely due to the high inertia of the ocean at these time scales.

[6] https://en.wikipedia.org/wiki/Boxcar_function

IV VALIDATION RESULTS

IV.1 Sea ice concentration

The quality of results for sea ice concentration is assessed by comparing with data from the Ocean and Sea Ice Satellite Application Facility Special Sensor Microwave Imager / Sounder (OSISAF SSMIS[7]) passive microwave data product, with a horizontal resolution of 10 km. The observations were first interpolated onto the model grid, and then compared with model results.

For the computation of bias and RMSD, the analysis was restricted to a "padded marginal ice zone" region, which was taken to be the area where the observed sea ice concentrations were in the range of [0.01, 0.9], supplemented by all grid cells within 200 km of this area. This algorithm was adopted to avoid artificial inflation of the quality assessment by excluding regions that are either in open water at a great distance from sea ice, or well within the nearly continuous complete ice cover in the sea ice's interior.

Estimated accuracy numbers and the time series of spatially integrated results are given in Table 6 and Figure 3, respectively.

Table 6: *Estimated accuracy numbers for sea ice concentration. A time-dependent region highlighting the transition from the open ocean to the ice covered interior was considered in the present analysis. The RMSD 90 percentile column gives the highest RMSD after discarding the largest 10% of daily values. Similarly, the 5 to 95 percentiles for bias were calculated. See Figure 1(a) for a detail on the regions.*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea ice concentration (area fraction)	**Full domain**	**SSMIS**	**0.13**	**0.16**	**0.04**	**[0.01, 0.08]**
	American s.		0.12	0.16	0.04	[0.01, 0.09]
	European s.		0.13	0.17	0.04	[0.00, 0.08]
	Asian s.		0.13	0.19	0.04	[-0,00, 0.10]

[7] https://osi-saf.eumetsat.int/products/osi-401-d

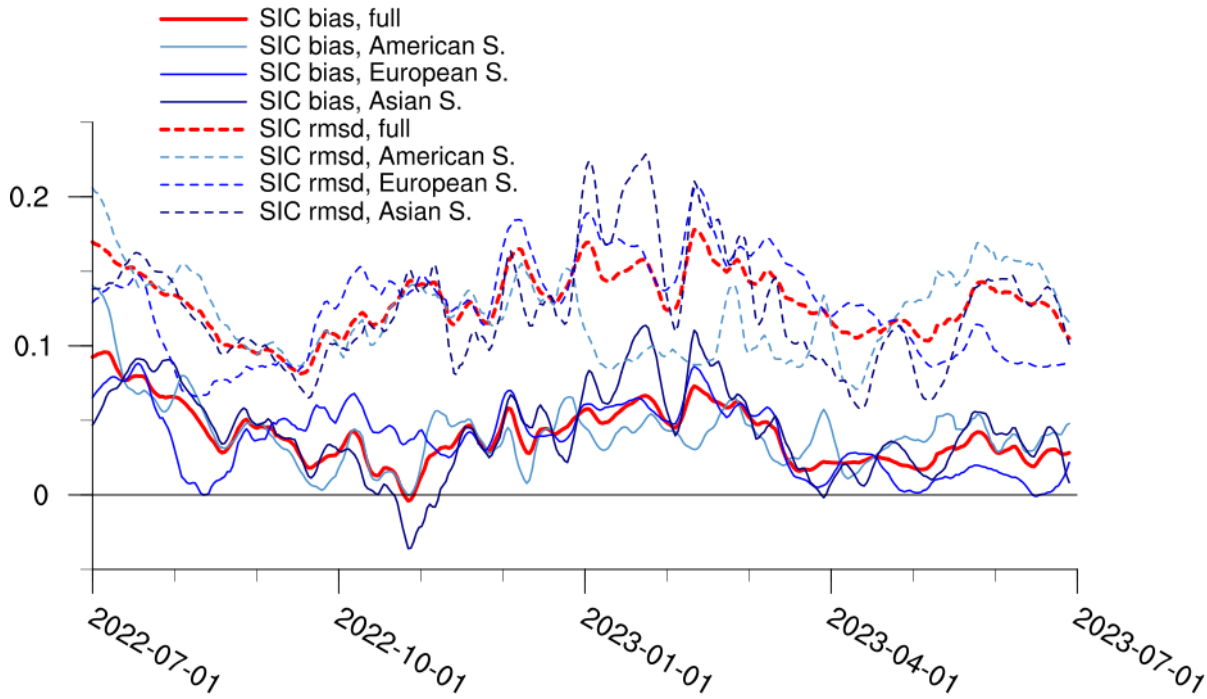


Figure 3: *Temporal evolution of sea ice concentration bias (full lines) and RMSD (dashed lines). Thick red lines show results for the entire Arctic Ocean. Thin lines in light, full and dark blue correspond to results for the American, European and Asian sectors, respectively. Regions are shown in Figure 1(a). Daily results are displayed, after smoothing with a seven-day boxcar filter.*

Table 7: *Contingency table for fractional distribution of sea ice concentration for observations versus model results. Sea ice concentration intervals that define the categories are in bold, while occupancy fractions are in regular font. See the text for details, including explanations of 'near' and 'far'. Numbers given in parenthesis indicate the total frequency that belonged to match-ups prior to filtering.*

		Observations							
	Sea ice concentration	**<0.01, far**	**<0.01, near**	**0.01-0.1**	**0.1-0.4**	**0.4-0.7**	**0.7-0.9**	**>0.9, near**	**>0.9, far**
Model results	**<0.01**	(0.448)	0.235	0.003	0.004	0.001	0.000	0	0
	0.01-0.1	0.002	0.044	0.003	0.009	0.002	0.000	0	0
	0.1-0.4	0.000	0.017	0.003	0.024	0.010	0.001	0.000	0
	0.4-0.7	0.000	0.003	0.001	0.014	0.035	0.006	0.000	0.000
	0.7-0.9	0.000	0.001	0.000	0.001	0.044	0.083	0.015	0.001
	>0.9	0.000	0.001	0.000	0.002	0.160	0.110	0.303	(0.182)

Model skill for the sea ice concentration is similar for the different regions (Table 6). Model bias is nearly always positive, with an excess in the sea ice concentration of around 0.05 (Figure 3). Furthermore, the offsets between observations and model results are largest in July and from December to March.

A match-up of model-observation concentration categories is provided in Table 7. The division of extreme categories into “near” and “far” relates to whether grid cells are located closer or further than 200 km from the concentration transition interval of [0.01, 0.9]. Thus, the matrix upper left and lower right corners are domains were any sea ice model should easily be able to reproduce observations. Accordingly, all model-observation match-ups that belong to these corners were removed before the fractional distribution was computed.

Moreover, adding up the diagonal numbers, we find that, on average, the model reproduces the observed categories in 68% of the region during this one-year period. In 26% of the cases, the model results are in the observations’ neighbour category. Finally, we find that the positive bias that we noted from Table 6 and Figure 3 is almost entirely due to somewhat high sea ice concentrations in the model when the observed concentrations are in the range [0.4, 0.9]. For this range the model concentration exceeds the observations by 0.08 (8%) on average.

Note that the data used for validation were also included in the model assimilation step.

IV.2 Sea ice edge

Sea ice edge metrics are examined using the same observational data as for sea ice concentration. Therefore, also in this case, observations were first interpolated onto the model grid, and then the comparison with model results was performed.

Two quantities are examined, the sea ice edge length and the distance between ice edge in the model results and the observed ice edge. In the present gridded products, the ice edge consists of the set of grid cells for which the sea ice concentration is 0.1 or larger, and at the same time has a neighbour grid cell with an ice concentration which Is lower than 0.1. The separation value of 0.1 was chosen since it coincides with one of the category separations applied for the contingency frequencies in Table 7 in Section IV.1.

Note that the validation data were derived from the OSISAF data (Table 5) which were used for assimilation of observed sea ice concentration (Table 4).

IV.2.1 Sea ice edge length

The model grid cells are squares with sides of length ds=6.25 km. Cells that belong to the ice edge usually have two neighbour edge grids. The contribution to the edge length in the direction of diagonal neighbours is set to $\sqrt{2}\cdot ds/2$. The contribution in the direction of other types of neighbour edge cells (right/left/up/down) Is set to $ds/2$.

As the sea ice edge length is different from other properties in this document (it has only one value for each region, and for each day), the bias simply becomes the difference between the modelled and observed values, and there is no RMSD for each day with results.

Table 8: *Estimated accuracy numbers for sea ice edge length. Values are in units of 1000 km (Kkm). The three sectors are shown in Figure 1(a). Results for RMSD are not available: as sea ice edge length has only one value for each region, and for each day the bias simply becomes the difference between the two values, and there is no RMSD. See the main text for the definition of edge length and for further details.*

Period: July 2022 – June 2023				
Variable	**Region**	**Supporting observations**	**Bias (model-observation)**	
			Mean	5-95%-ile range
Sea ice edge length (Kkm)	**Full domain**	**SSMIS**	**0.11**	**[-1.5, 1.5]**
	American s.		-0.01	[-1.3, 1.2]
	European s.		0.13	[-0.6, 1.1]
	Asian s.		-0.01	[-1,3, 1.2]

For reference, the average ice edge length in the observations is 12.1 Kkm, with 90% of the length data in the range [9.3 Kkm, 17.6 Kkm].

We note from Table 8 that the mean bias is slightly positive, and on average due to (small) offsets in the European sector. The bias is likely due to either a smoother edge in the observations, and/or a shift in the position, as edge lengths will be affected by the representation of the sea ice extent.

The temporal evolution of the sea ice edge length bias is displayed in Figure 4. We observe that, while the time mean values are close to 0, there is a large variability for the 12 months that are analysed, with no obvious seasonal signal. The biases in the three subregions are largely uncorrelated, with correlations in the range ±0.3.

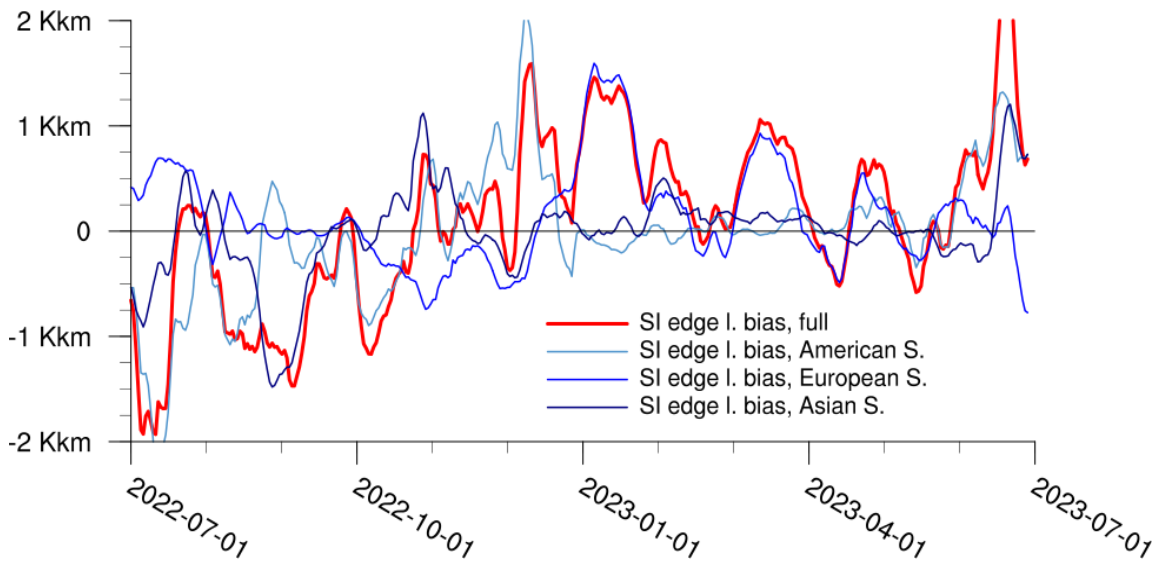


Figure 4: *Temporal evolution of bias for ice edge length. Regions are shown in Figure 1(a). A seven-day boxcar filter was applied.*

IV.2.2 Sea ice edge position

Model skill for predicting ice edge position is examined through the closest distance between the model and observational grid cells containing the ice edge. We take this distance to be positive if the model ice edge "overshoots" the observational edge (on the open ocean side) and vice versa.

Then, the algorithm is re-run, but with observation edge grid cells as the starting point. Subsequently, the edge distance in the analysis is taken to be the average of the two. This algorithm is explained in detail in Melsom et al., (2019), and we are presenting results that correspond to D^{IE}_{AVG} and D^{IE}_{RMS} in their article.

Table 9: *Estimated accuracy numbers for sea ice edge position (units: km). See Section III.3 for an explanation of the percentiles. The three sectors are shown in Figure 1(a).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea ice edge position (km)	**Full domain**	**SSMIS**	**78**	**112**	**1**	**[-25, 41]**
	American s.		88	146	-3	[-67, 58]
	European s.		56	91	2	[-19, 26]
	Asian s.		65	104	10	[-33, 61]

The temporal evolution of these two metrics is displayed in Figure 5. We note that RMSD tends to be highest in the American sector. This observation is confirmed by the results in Table 9.

In addition, the bias becomes negative after the sea ice minimum in mid-September.

IV.3 Sea ice thickness

Remote sensing data for sea ice thickness utilizes various instruments and analyses. Data from the thickness of thin sea ice (up to approximately 1 m) can be derived from SMOS satellite observations, while data for the thickness of thick sea ice are available from processing of altimeter data. Due to the orbital parameters, the altimeter on Cryosat-2 is particularly useful.

We will separate the validation of sea ice thickness for thin sea ice and thick sea ice according to the availability of data from these two sources. Note that due to changing conditions from summer to winter. sea ice thickness data is only available during a period that roughly corresponds to the time of year when freezing dominates over melting.

Note that data from a SMOS-Cryosat combination product are used for assimilation.

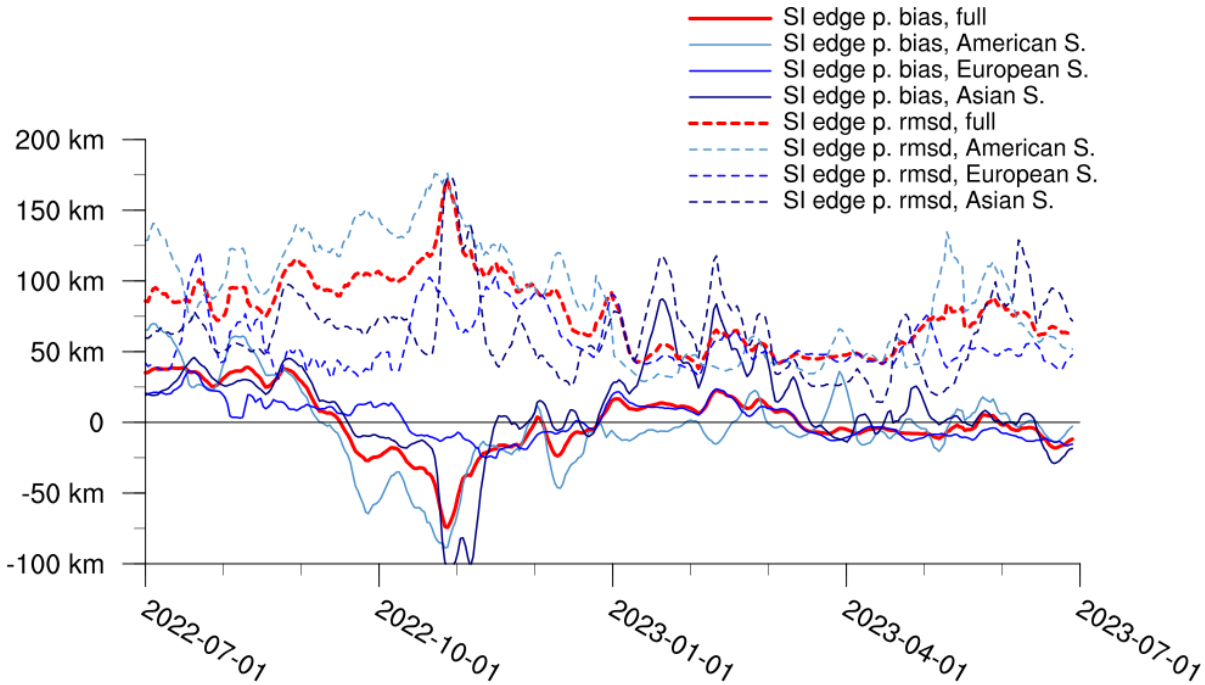


Figure 5: *Temporal evolution of sea ice edge position bias (full lines) and RMSD (dashed lines). Distances are positive when the model ice edge "overshoots" the observational edge. Thick red lines represent the entire Arctic Ocean. Sectors are shown in Figure 1(a). Daily results, after smoothing with a seven-day boxcar filter.*

IV.3.1 Thickness of thin sea ice

The quality of results for the thickness of thin sea ice is assessed by comparing model results with data from the Soil Moisture and Ocean Salinity (SMOS) product. SMOS results and uncertainties are available from the Alfred Wegener Institute. The processing of the remote sensing data has been described by Kaleschke et al., (2016). Daily data are available for the period from October to April, with a horizontal resolution of 12.5 km. Observations with uncertainties in excess of 1 m were discarded, as was data which had uncertainties exceeding 75% of the thickness value.

The observations were first interpolated onto the model grid prior to comparing with model results. Resulting estimated accuracy numbers for the thickness of thin (<1 m) sea ice are given in Table 10.

Table 10: *Estimated accuracy numbers for the thickness of thin sea ice. Units: m. See Section III.3 for an explanation of the percentiles.*

Period: October 2022 - April 2024						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea ice thickness (m)	**Full domain**	**SMOS**	**0.33**	**0.45**	**0.11**	**[-0.02, 0.26]**

Table 11: *Contingency table for fractional distribution of thickness of thin sea ice, for observations versus model results. Categories defined by thickness intervals are in bold, with values in m. Occupancy fractions are in regular font. Number in parenthesis indicates the total frequency that belonged to the 0-0 match-up (prior to filtering, see the main text for further details)*

		Observations				
	Sea ice thickness (m)	**0**	**0-0.3**	**0.3-0.6**	**0.6-1**	**>1**
Model results	**0**	(0.711)	0.057	0.000	0.000	0
	0-0.3	0.071	0.226	0.039	0.007	0
	0.3-0.6	0.016	0.078	0.113	0.067	0
	0.6-1	0.013	0.027	0.067	0.107	0
	>1	0.016	0.010	0.025	0.061	0

A match-up of model-observation thickness categories is provided in Table 11. The observed domain includes regions that are separated from the ice cover by some distance, and consequently have thicknesses of 0 m. Accordingly, all model-observation match-ups that belong to the upper left corner were removed before the fractional distribution was computed. The number given in parenthesis in the table corner indicates the total frequency that belonged to the 0-0 match-up (prior to the filtering).

Adding up the numbers on the diagonal entries in Table 11, we find that the model on average reproduces the observed categories in 45% of the ice-covered region during the half year period with data. In 44% of the cases the model results are in the observations' neighbour category. Finally, we note that the positive bias in Table 10 is reflected in all thickness categories.

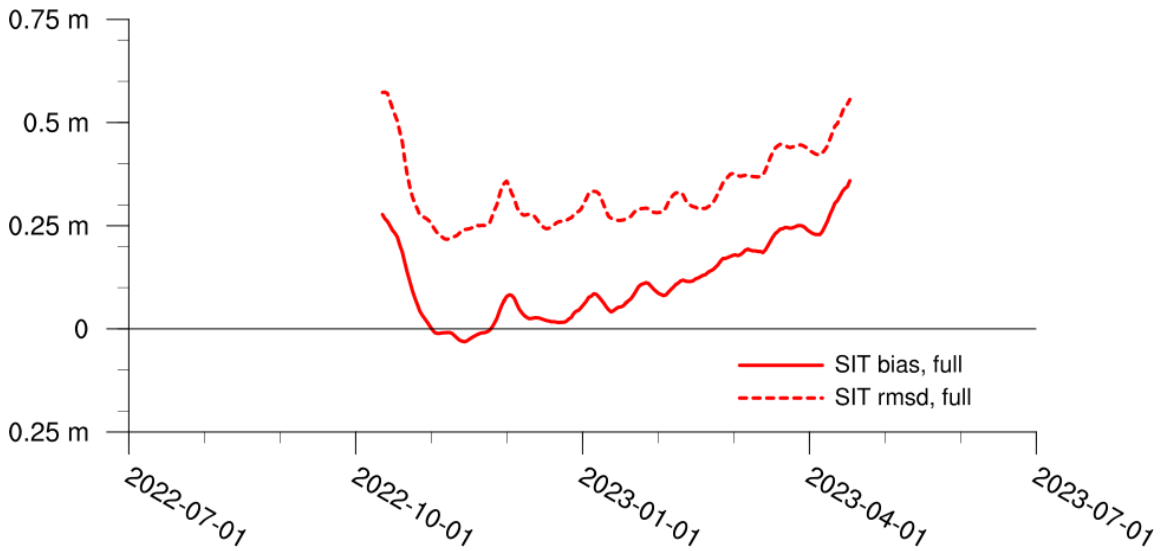


Figure 6: *Temporal evolution of the bias (solid line) and RMSD (dashed line) for the thickness of thin sea ice. Daily results, after application of a seven-day boxcar filter.*

The temporal evolution of the bias and RMSD Is displayed in Figure 6. We note that the model discrepancy is relatively large in the beginning of the period, and increasing towards the end, with moderate differences in the intermediary period. Since the model assimilates a product that incorporates the SMOS data, the initial drop in the discrepancy may be related to assimilation.

Table 12: *Estimated accuracy numbers for the thickness of thick sea ice. Values are given in units of m. See Section III.3 for an explanation of the percentiles.*

Period: October 2022 – March 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea ice thickness (m)	**Full domain**	**Cryosat-SIRAL**	**0.78**	**0.94**	**-0.31**	**[-0.59, 0.24]**

IV.3.2 Thickness of thick sea ice

The quality of results for the thickness of thick sea ice is assessed by comparing with data from the SAR Interferometric Radar Altimeter (SIRAL) instrument on the Cryosat-2 satellite (Laxon et al., 2014). The data, which comes on a horizontal resolution of 5 km, are distributed by the Centre for Polar Observation and Modelling (CPOM[8]). This dataset also includes an error estimate given by the observations' spatially varying standard deviation uncertainties. We have here used the biweekly dataset version. Data are available for the period from October 2022 to March 2023.

Table 13: *Contingency table for fractional distribution of thickness of thick sea ice, for observations versus model results. Categories defined by thickness intervals are in bold, with values in m. Occupancy fractions are in regular font. See the main text for details.*

		Observations						
	Sea ice thickness (m)	**0**	**0-1**	**1-1.5**	**1.5-2**	**2-2.5**	**2.5-3**	**>3**
Model results	**0**	0	0.000	0.000	0.000	0	0	0
	0-1	0	0.034	0.174	0.098	0.019	0.005	0.003
	1-1.5	0	0.004	0.083	0.162	0.037	0.005	0.002
	1.5-2	0	0.002	0.024	0.085	0.070	0.011	0.003
	2-2.5	0	0.002	0.013	0.029	0.050	0.021	0.004
	2.5-3	0	0.001	0.005	0.006	0.005	0.013	0.013
	>3	0	0.001	0.005	0.005	0.001	0.001	0.004

From Table 12 we find that the thickness of thick sea ice is generally somewhat smaller in the model than in the observations derived from Cryosat-2 data.

A match-up of model-observation thickness categories are provided in Table 13. Categories are mostly defined with thickness intervals of 0.5 m, and occupancy frequencies are tabulated.

The sum of diagonal entries in Table 13 reveal that the model on average reproduces the observed categories in 27% of the ice-covered region during the half year period. In 50% of the cases, the model results are in the observations' neighbour category, most frequently shifted by one category in the direction of thicker sea ice in the observations. This reflects the negative thickness bias that was reported in Table 12.

Finally, we take advantage of the quantification of the standard deviations of the Cryosat-2 thickness product. An alternative assessment of the quality is then achieved by examining which model results that are locally within ±1 standard deviation of the observed value.

The results of such an examination is presented in Figure 7. We note that generally, the model results are within ±1 standard deviation of the observations. Around 20% of the model values for thick sea ice

[8] https://cpom.org.uk/

are lower than the observations minus one standard deviation. With the exception of October, model results are almost never above the observational value plus one standard deviation.

Finally, since the model bias for thin sea ice is positive (0.11m; Table 10), while it is negative for thick sea ice (-0.31m, Table 12), we conclude that the model sea ice thickness is too smooth.

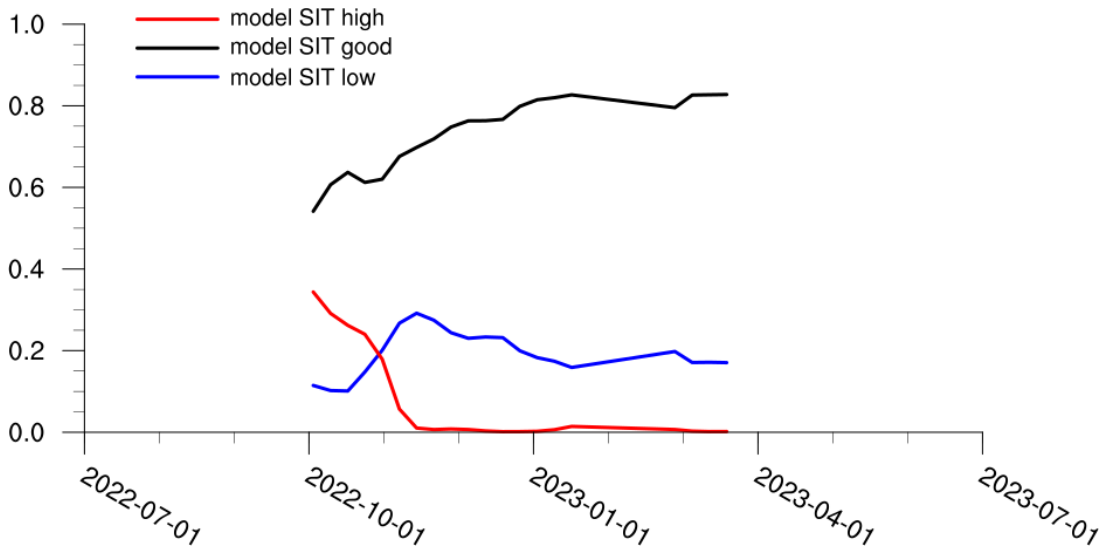


Figure 7: *Temporal evolution of the differences of model results and Cryosat-2 results for the thickness of thick ice. The black curve shows the fraction of model grid cells which is inside the range of the observed value ±1 standard deviation. Red and blue curves correspond to the fraction of cells with thicknesses that are larger than the observations plus one standard deviation, and smaller than the observations minus one standard deviation, respectively. Results are produced with a weekly resolution.*

IV.4 Sea ice drift

We compare model results for sea ice drift with sea ice displacement vector data based on SAR observations that are provided in the Copernicus Marine Service catalogue by the Danish Technical University (DTU). These data are not assimilated in the model system. Ice drift from another remote sensing processing is assimilated. The two datasets are constructed using different sensors that measures radiation on non-overlapping frequency ranges, and are thus datasets that are independent.

The observational product computes displacement vectors using Maximum Cross Correlation (MCC) processing of overlapping SAR data. Observations are available for periods that span 23-25 hours, sampled irregularly during each day. Prior to the analysis here, the displacement vector distances are scaled to represent 24 h drift. Then the observations are mapped from a 10 km grid onto the map projection and resolution for which model results are available.

Corresponding model displacements are derived from integration of the sea ice velocity fields. Note that the model results do not include tides. However, tidal effects are modest in the observations, due to the 23-25 h window that is used.

The displacement that we consider here is a vector property. The definition of "bias" and "RMSD" must then be modified. In the present examination, the algorithm that was explained in Section III.3 and illustrated in Figure 2, was applied.

Estimated accuracy numbers that resulted from such an analysis are listed in Table 14 below. We note that the bias in the length of the drift displacement vectors are small, e.g. compared with RMS of the separation distance between vector endpoints. Hence, the main contrasts between the observed and modelled sea ice drift is for the direction, and not the distance.

If we let the listed metrics values in Table 14 represent a single vector, and note that the mean length of the observed displacement vectors is 7.5 km, we find that the directional offset between observations and model displacement vector end positions would be 42° in the full domain case.

***Table 14**: Estimated accuracy numbers for sea ice displacement vectors due to 24 h drift. The results are based on averages from a one-year period with forecasts. The three sectors for the region are shown in Figure 1(a).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea ice drift displacement (km)	**Full domain**	**SAR**	**5.4**	**6.9**	**-0.2**	**[-3.2, 2.9]**
	American s.		4.2	6.0	0.7	[-2.6, 3.8]
	European s.		6.7	10.4	-1.1	[-6.2, 3.4]
	Asian s.		5.7	9.2	-1.3	[-7.2, 3.6]

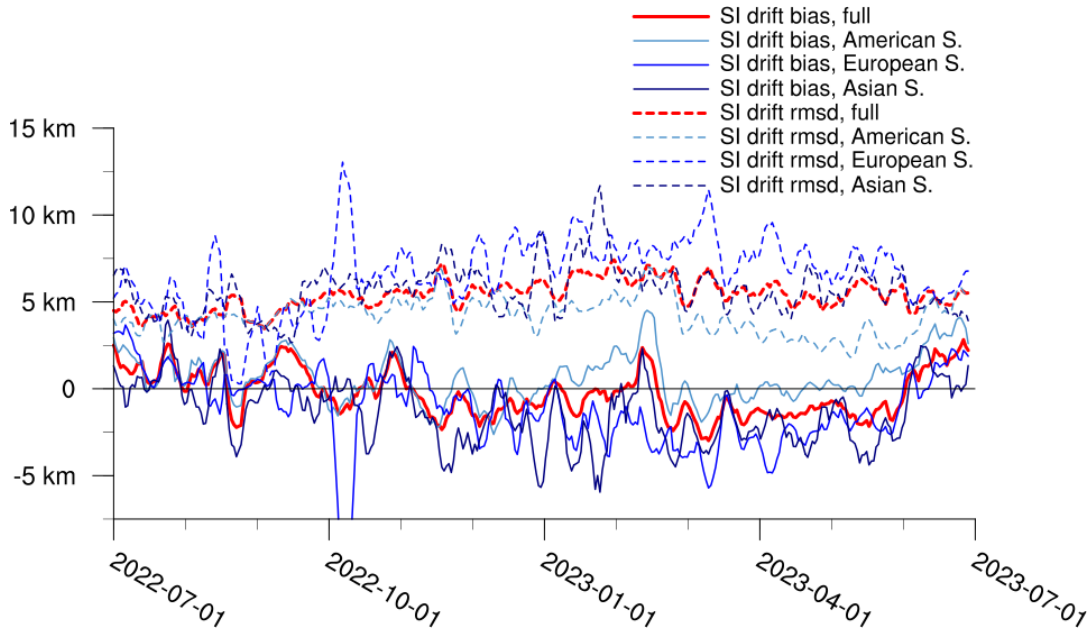


***Figure 8**: Temporal evolution of sea ice drift bias (solid lines) and RMSD (dashed lines), as defined in Figure 2. The bias values were computed by subtracting observed 24 h drift distances from the corresponding model results. Thick red lines show results when the entire Arctic Ocean is considered. Lines in shades of blue are for three sectors which are separated by longitude, as displayed in Figure 1(a). Displayed here are daily results, after smoothing with a seven-day boxcar filter.*

Results for the two metrics and the four regions are presented as daily time series in Figure 8. We find that while there is somewhat larger temporal variability for the three sub-regions than for the full domain, there is no obvious seasonal variation in the results for these metrics.

IV.5 Ocean drift

For the analysis of the quality of model results for ocean drift, we consider derivations from two sources of observational data: trajectories of surface drifting buoys, and the surface slope along altimeter ground tracks.

IV.5.1 Near-surface drift

Model results for the near-surface drift have been compared with position data from drifting buoys. The buoys have a drogue that is centred on a depth of 15 m. A surface sensor records the sea surface temperature, which is transmitted along with the buoy's position to satellites above.

The data from the Surface Velocity Program (SVP) drifters (Lopez et al., 2014) are compiled as part of the Global Drifter Program, which is managed by NOAA[9]. The data are redistributed by the Copernicus Marine Service. The trajectories of the buoys from which data are available during the period that is examined here (2022-07-01 – 2023-06-30) are displayed in Figure 9.

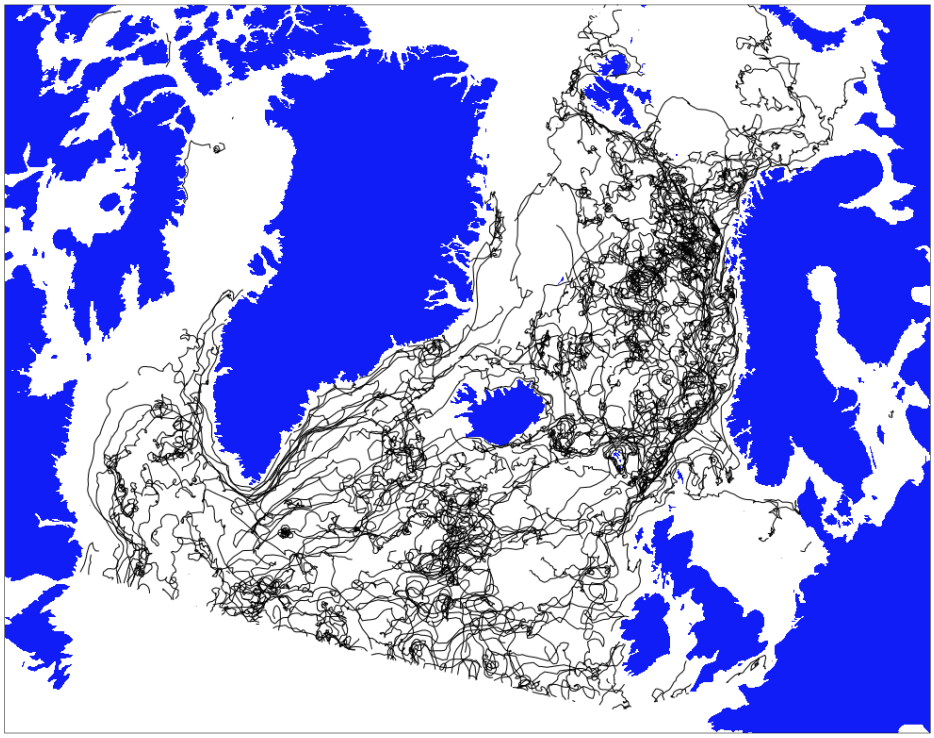


***Figure 9**: Trajectories of the 243 surface drifters from which position data and SST observations were used in the present analysis. Trajectories to the north of 50°N are displayed. The data are from the period 2022-07-01 – 2023-06-30.*

[9] https://www.aoml.noaa.gov/global-drifter-program/

We compare drift using the model results for horizontal velocity components at 15 m to simulate the drift of floats, using the SVP drifters' positions at 00 UTC as the starting points for the integration. This gives us simulated trajectory end positions after 24 h drift. We then proceed completely analogously to the analysis of sea ice drift, as described in Section IV.4, and present results for the same metrics. Note that drift distance is the separation of the end position from the starting position, and not the length of the trajectory.

***Table 15**: Estimated accuracy numbers for near-surface drift distance. Results are given for 24 h drift, for the trajectories displayed in Figure 9. Metrics for vector quantities are defined in Section III.3, where an explanation of the percentiles is given. The domains of the Nordic Seas subregion and the Barents Sea subregion are shown in Figure 1(b).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Near-surface drift distance (km)	**Full domain**	**SVP drifters**	**18**	**22**	**-8**	**[-13, -5]**
	Nordic Seas		17	22	-9	[-13, -5]
	Barents Sea		14	22	-8	[-18, -0]

The results for the estimated accuracy numbers are given in Table 15. We note that the model consistently underestimates the observed drift. The overall mean 24 h drift distance for all observed trajectories is 16 km, which gives an average underestimation of 50%. The root mean squares of the distances between the end positions of simulated and observed 24 h drift are nearly equal, so the inconsistency of the model results and observations is also substantial for the drift direction.

The time series of mean daily values for the metrics are displayed in Figure 10. We see that there is little seasonal variability in the quality of the model. The large variability in the results for the Barents Sea can be attributed to the low number of drifters in that region.

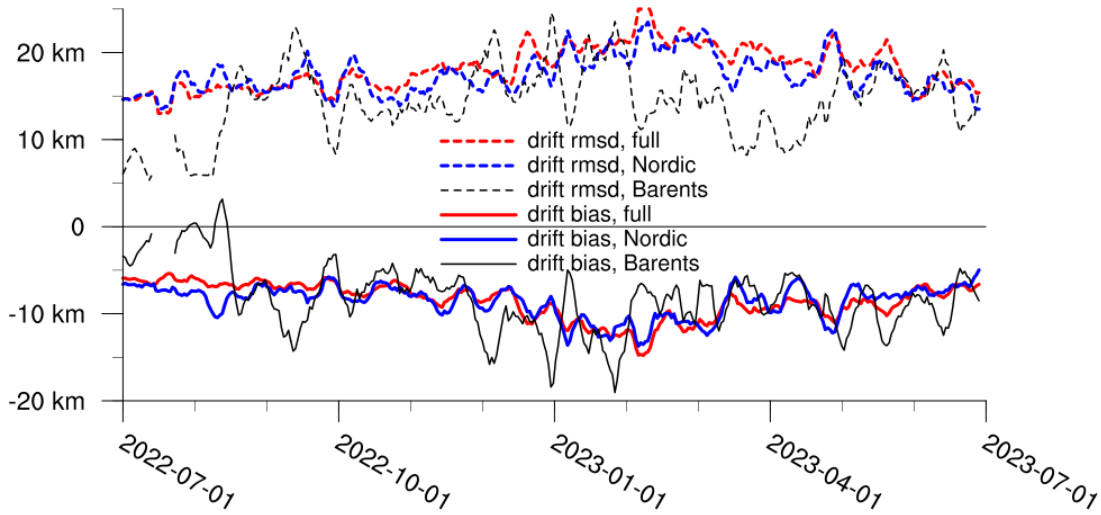


***Figure 10**: Temporal evolution of near-surface drift distance bias (solid lines) and RMSD of end position separation (dashed lines). The bias values were computed by subtracting observed 24 h drift distances from the corresponding model results. Thick red lines show results for the entire Arctic Ocean. Daily results, after smoothing with a seven-day boxcar filter.*

IV.5.2 Surface geostrophic velocity

At some distance from the equator, the horizontal pressure gradient and the Coriolis effect are usually the leading order balance in the horizontal momentum equation. The resulting current is referred to as the geostrophic velocity.

The surface geostrophic velocity component in the direction perpendicular to satellite ground tracks can be derived from the slope of the measured sea level which is derived from altimetry data. Here, we compare this velocity component to computations based on the sea level slope in the model results, extracted along the same ground tracks.

***Table 16**: Estimated accuracy numbers for component of the surface geostrophic velocity perpendicular to ground tracks of selected satellites. Units: cm/s. See Section III.3 for an explanation of the percentiles. The Nordic Seas domain is shown in Figure 1(b).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Surface geostrophic velocity component (cm/s)	**Full domain**	**Sentinel-6a**	**10**	**11**	**-6**	**[-7, -5]**
		Jason-3	**9**	**11**	**-5**	**[-7, -5]**
		Cryosat-2	**10**	**11**	**-5**	**[-7. -4]**
	Nordic Seas	Sentinel-6a	8	10	-5	[-7, -3]
		Jason-3	9	11	-5	[-7, -3]
		Cryosat-2	10	12	-7	[-9, -5]

We take the velocity component derived from altimetry data to be positive. If the sign of the slope is different in model results and observations, the model velocity component becomes negative. Note also that in contrast to the analysis in Section IV.5.1, the quantity we examine here is scalar.

As with the results from 24 h drift of surface buoys, the model consistently underestimates the strength of the velocity component under consideration. If we take a velocity bias of 5 cm/s and assume that it is constant, this corresponds to a distance of 4.3 km in the span of 24 h. If we further assume that the direction of the 2D bias vector is independent of the satellite ground tracks, the distance bias becomes 6.1 km. This is similar to, but slightly lower than, the drift distance bias we inferred from position data from drifting buoys.

The time series of mean daily values for the bias are displayed in Figure 11. The seasonal variability for this quality metric is low.

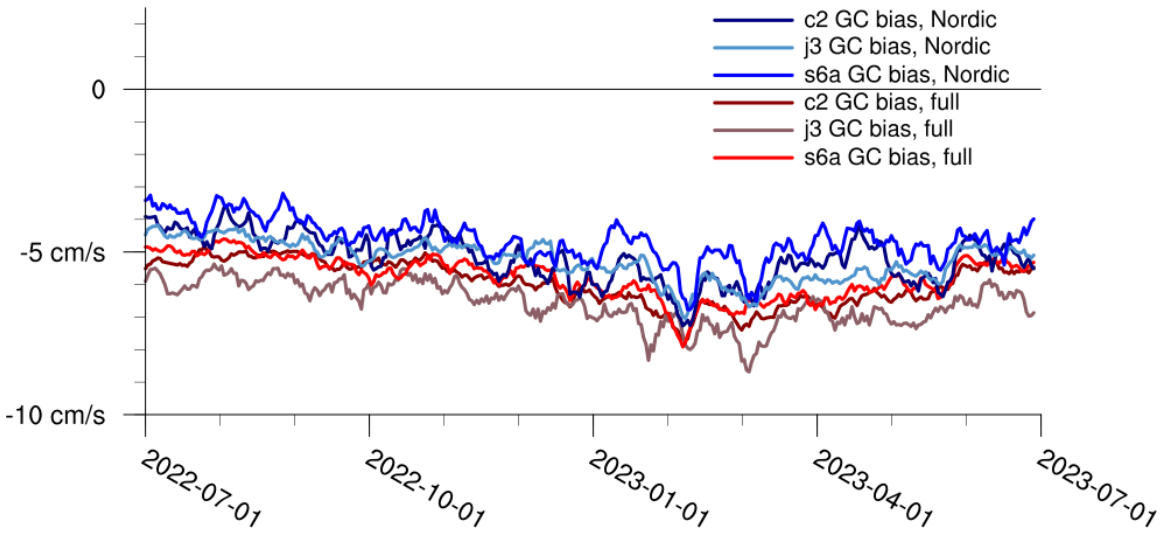


Figure 11: *Temporal evolution for the bias of the geostrophic current component perpendicular to satellite ground tracks. Results are derived by comparing with altimetry data from Sentinel-6a, Jason-3 and Cryosat-2. Results for the full domain are shown in red and brown, while corresponding results for the Nordic Seas are in blue. Daily results, after smoothing with a seven-day boxcar filter.*

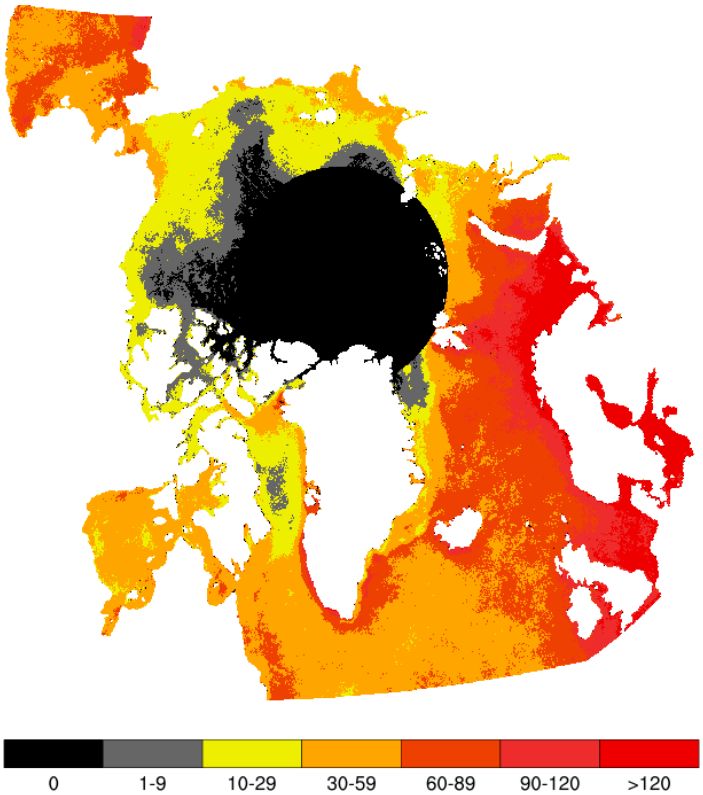


***Figure 12**: Number of days for which retrieved AVHHR Metop-B SST data was of sufficient quality for the period 2022-07-01 – 2023-06-30.*

IV.6 Temperature

The spatio-temporal coverage of sea surface temperature (SST) observations is much better than for vertical temperature profiles: For the 1 year period starting on 2022-07-01, there are 3000 vertical profiles, while the accumulated number of grid cells with Metop-B SST data is $25 \cdot 10^6$. The data sources also differ between SST and vertical profiles. Consequently, model skill is assessed separately for SST and vertical temperature profiles.

***Table 17**: Estimated accuracy numbers for sea surface temperature. Results are given in units of K. See Section III.3 for an explanation of the percentiles. The domains of the Nordic Seas and the Barents Sea subregions are shown in Figure 1(b).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea surface temperature (K)	**Full domain**	**SVP drifters**	**0.7**	**0.9**	**0.1**	**[-0.1, 0.3]**
		AVHRR	**0.8**	**1.3**	**0.1**	**[-0.2, 0.3]**
	Nordic Seas	SVP drifters	0.9	1.2	0.1	[-0.1, 0.5]
		AVHRR	0.7	1.0	0.1	[-0.3, 0.5]
	Barents Sea	SVP drifters	0.6	1.0	0.0	[-0.4, 0.6]
		AVHRR	0.7	1.1	0.3	[-0.0, 0.7]

IV.6.1 Sea surface temperature

For surface temperature, data are available from both in situ and remote sensing satellite-borne instruments. Here, we present results for quality metrics that are based on in situ data from Surface Velocity Program (SVP) drifters, and temperatures derived from infrared AVHRR observations (remote sensing) from the Metop-B satellite. The in-situ sea surface temperature data are not assimilated by the model, whereas the AVHRR data are one of many sources for the product which is used for assimilation.

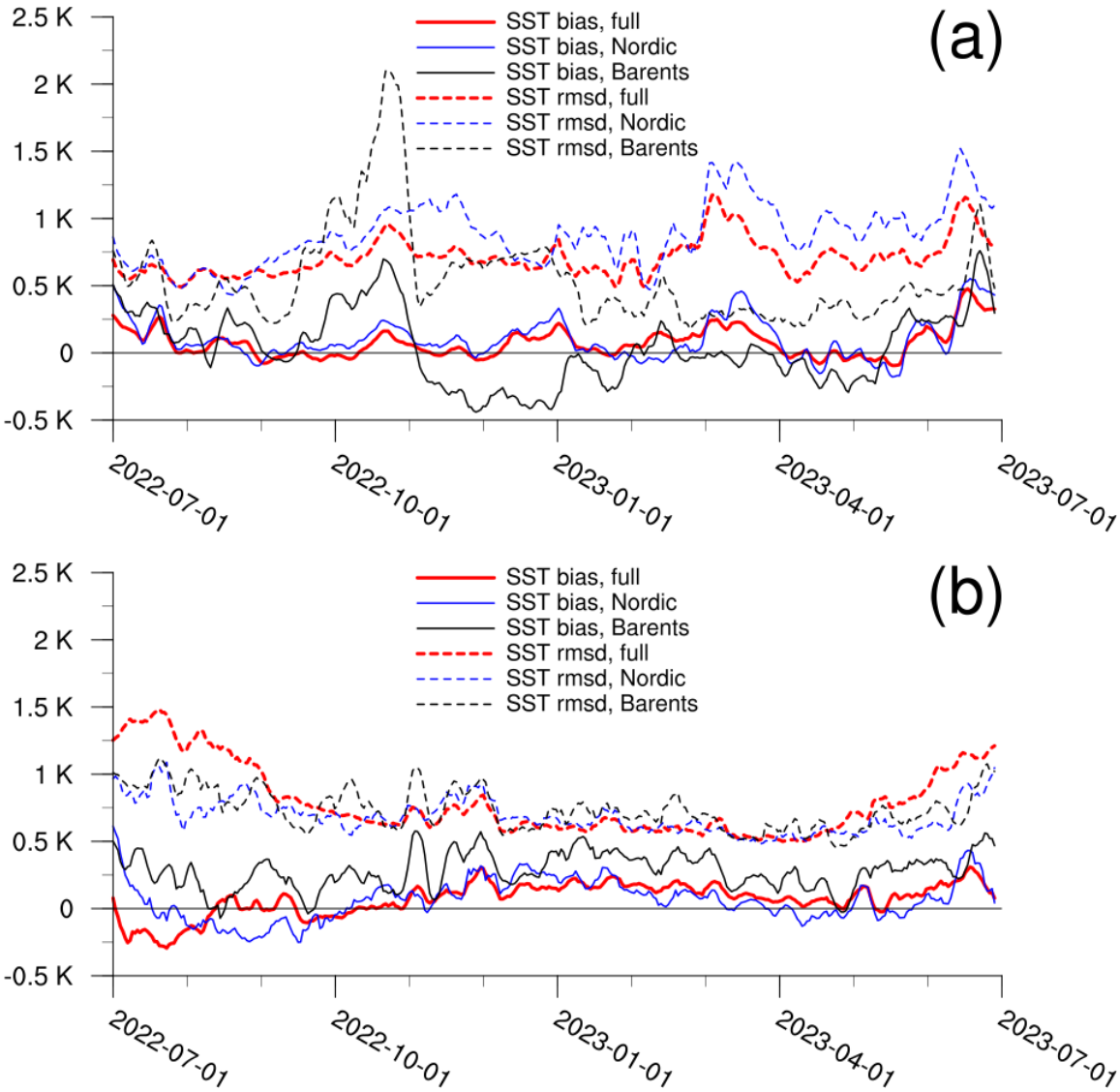


Figure 13: *Temporal evolution for the bias and RMSD of the observed and modelled sea surface temperature results for the full domain and sub-regions, based on comparisons with (a) surface drifters data and (b) data derived from AVHRR infrared observations. Daily results, after smoothing with a seven-day boxcar filter.*

Trajectories of the drifters that provide the in-situ data are shown in Figure 9.

Infrared observations of sea surface temperature are only available under cloud-free conditions, and hence the data coverage varies spatially and seasonally. The spatial distribution of number of days with data of an acceptable quality is displayed in Figure 12.

We find that the quality metrics have very similar values irrespective of the information source and the region. The model results have a positive sea surface temperature bias of about 0.1 K, which is indicative of high quality. The root mean square temperature differences are on average below 1 K, which also reflects the high quality of the model results for sea surface temperature.

The time series of mean daily values for the metrics are displayed in Figure 13. We see that there is little seasonal variability in the quality of the model.

When the drift distance was compared with position data for the same buoys in Section IV.5.1, we found a notable variability in the quality metrics for the Barents Sea, which was attributed to the modest sampling in that region. We see from Figure 13(a) that such a variability is even more pronounced for the sea surface temperature observations.

IV.6.2 Potential temperature in vertical layers

We assess the quality of the model results for potential temperature in a set of five vertical layers. The depth span of each of these layers are listed in Table 18. The averaged layer values after vertical integration were used for this purpose. Profiles without data at depths beneath 200 m were discarded.

Table 18: *Estimated accuracy numbers for potential temperature, based on averages for five layers. Values are in units of K. See Section III.3 for an explanation of the percentiles*

Period: July 2022 – June 2023							
Variable	**Region**	**Supporting observations**	**Layer**	**RMS difference**		**Bias (model-observation)**	
				Mean	90%-ile	**Mean**	5-95%-ile range
Potential temperature (K)	**Full domain**	**Profiling instruments**	**0-5**	**0.35**	**0.62**	**-0.03**	**[-0.39, 0.30]**
			5-100	**0.37**	**0.64**	**-0.10**	**[-0.41, 0.18]**
			100-300	**0.41**	**0.73**	**0.11**	**[-0.18, 0.47]**
			300-800	**0.56**	**1.00**	**0.17**	**[-0.26, 0.64]**
			800-2000	**0.36**	**0.64**	**-0.15**	**[-0.46, 0.15]**
	Nordic Seas		0-5	0.33	0.64	-0.04	[-0.48, 0.38]
			5-100	0.39	0.76	-0.15	[-0.53, 0.19]
			100-300	0.42	0.80	0.06	[-0.31, 0.57]
			300-800	0.43	0.82	0.01	[-0.43, 0.45]
			800-2000	0.27	0.40	-0.22	[-0.40, 0.00]

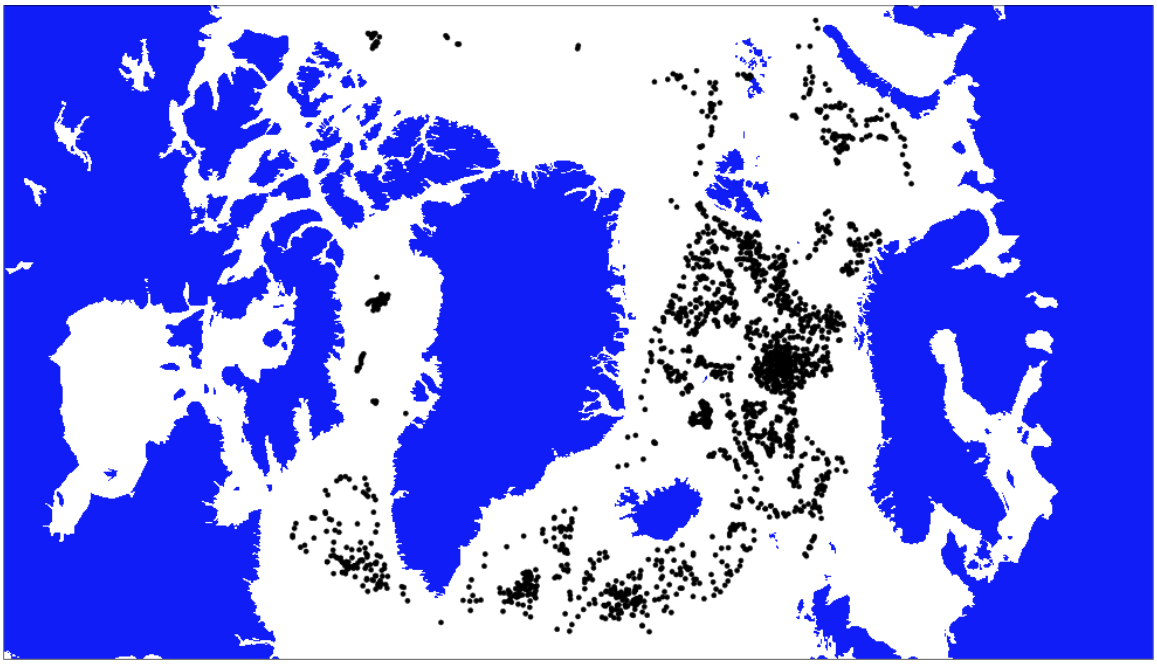


***Figure 14**: Locations of the 2976 profiles from which temperature and salinity observations were available for the period 2022-07-01 – 2023-06-30.The location of the analysed profiles, north of 60°N, are displayed.*

A total of 2976 profiles of vertical temperature were available for this examination. ARGO buoys were the dominating source for these observations. The positions of these profiles are shown in Figure 14. Not all profiles include data from all of the five layers. The number of available profiles, in italics, from the five layers defined in Table 18, have this distribution: Layer 1, 0-5 m: *1906*; layer 2, 5-100 m: *2937*; layer 3, 100-300 m: *2976*; layer 4, 300-800 m: *2741*; layer 5, 800-2000 m: *2046*.

From Table 18 we find that the quality of the model results is fairly uniform in the vertical dimension, and differences between the results for the Nordic Seas and for the full domain are small. We also note that temperature biases are nearly the same as we found for the sea surface temperature in the previous sub-section.

The bias and RMSD time series are displayed in Figure 15 and Figure 16, respectively. Panels (a) and (b) show results for the full domain and for the Nordic Seas sub-domain, respectively. Note that the application of a seven-day boxcar filter in these time series strongly influences the representation of the metrics, particularly the RMSDs in the upper layers. This is primarily due to the small sample for each day, which is typically 5-10 profiles, sometimes even fewer.

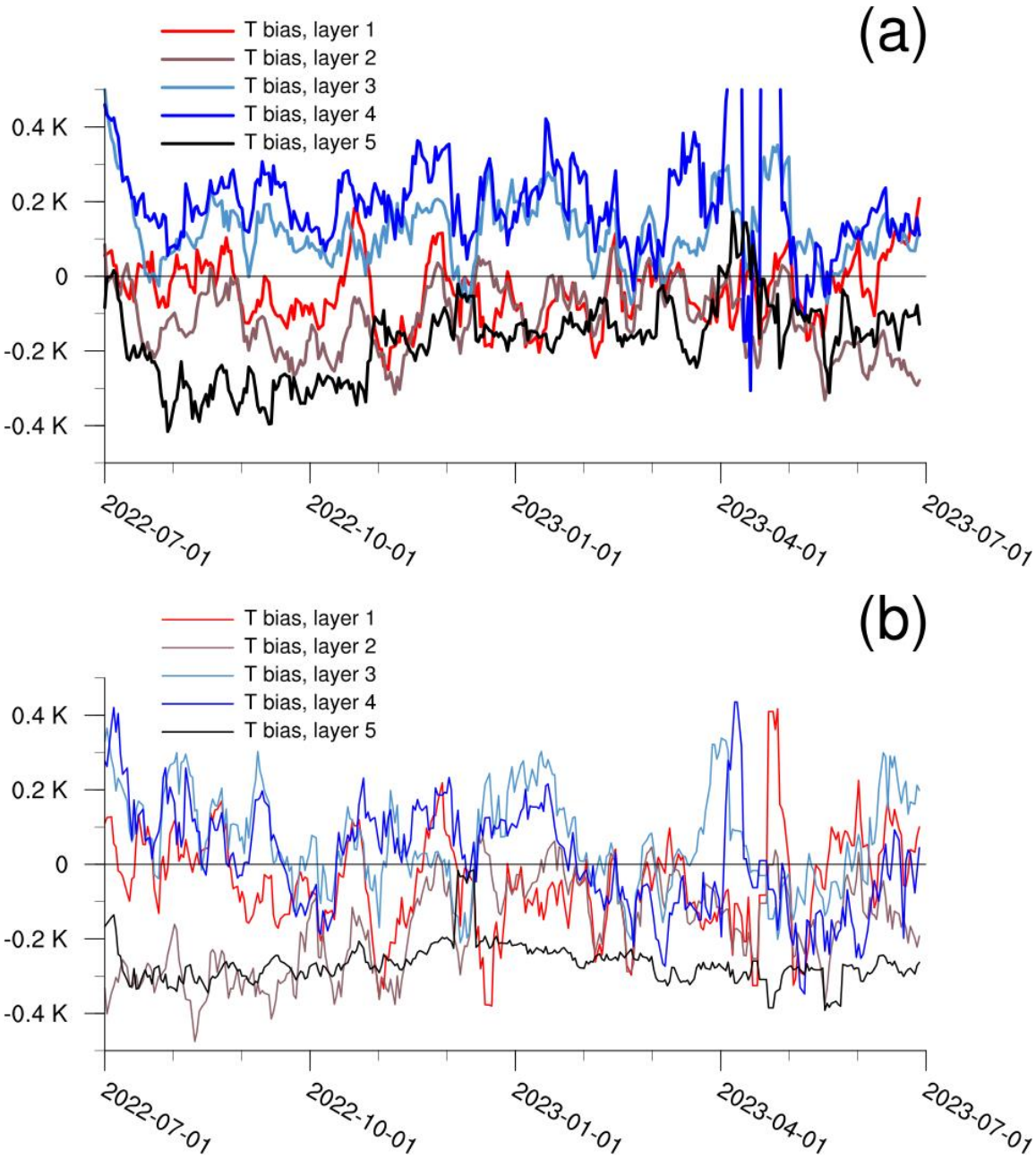


Figure 15: *Temporal evolution of the bias of the model results for potential temperature, for the five layers in the present analysis for (a) the full domain and (b) the Nordic Seas sub-domain. Displayed here are daily results, after smoothing with a seven-day boxcar filter.*

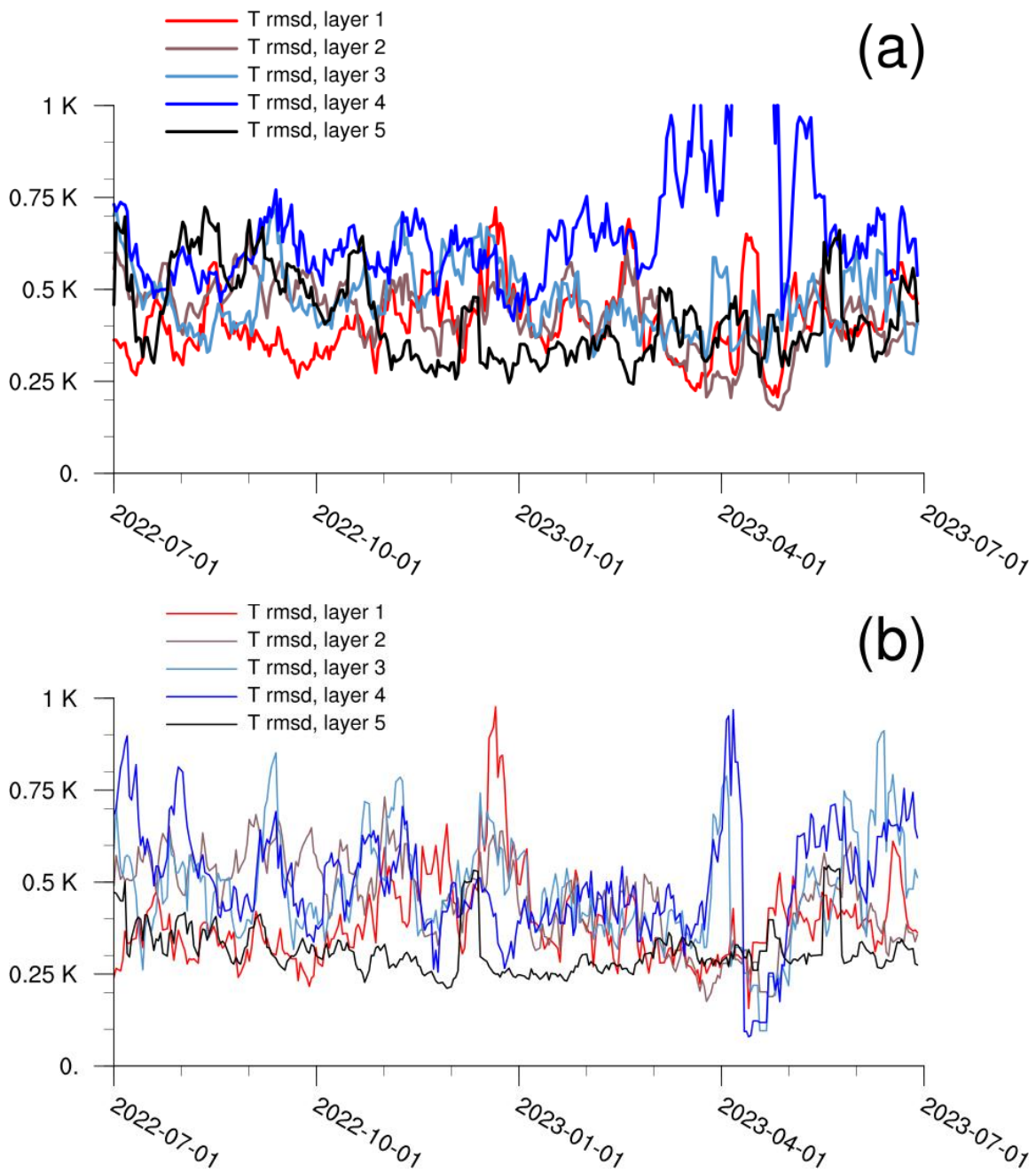


Figure 16: *Temporal evolution of the RMSD of the model results for potential temperature, for the five layers in the present analysis for (a) the full domain and (b) the Nordic Seas sub-domain. Daily results, after smoothing with a seven-day boxcar filter.*

IV.7 Salinity

Salinity is analysed in the same manner as potential temperature (Section IV.6.2).

We use the same observational dataset and process model results and observations following the same computational algorithm, to determine model skill for five vertical layers, for the full domain, and for the Nordic Seas sub-domain. The amount of data is identical to the availability of data for temperature in Section IV.6.2 (2976 profiles). The distribution of the data is as shown in Figure 14.

The EANs for salinity indicate that the quality of the model results is high (Table 19). Biases are near 0 and non-systematic. We also note that the EAN values decrease with increasing depth.

Bias and RMSD time series are displayed in Figure 17 and Figure 18, respectively. Note that the application of a seven-day boxcar filter in these time series strongly influences the representation of the metrics, particularly the RMSDs in the upper layers. This is primarily due to the small number of samples for each day, typically 5-10 profiles, sometimes even fewer.

We note that the metrics have markedly larger variability in the upper two layers than below. Furthermore, there is a tendency of improved quality during late winter and early spring.

We note with some concern that the salinity bias in the full domain case increases in layers 2 and 3 towards the end of the time series; see Figure 17(b). In a supplementary analysis, results from the two months that follow (July and August, 2023) were added to the examination. Fortunately, we find that these biases are reduced during the extension.

Table 19: *Estimated accuracy numbers for salinity, based on averages for five layers. Values are in units of ‰ (10^{-3}).*

Period: July 2022 – June 2023							
Variable	**Region**	**Supporting observations**	**Layer**	**RMS difference**		**Bias (model-observation)**	
				Mean	90%-ile	**Mean**	5-95%-ile range
Salinity (‰)	**Full domain**	**Profiling instruments**	**0-5**	**0.11**	**0.25**	**-0.01**	**[-0.11, 0.11]**
			5-100	**0.09**	**0.17**	**-0.01**	**[-0.09, 0.09]**
			100-300	**0.05**	**0.09**	**-0.00**	**[-0.04, 0.05]**
			300-800	**0.03**	**0.05**	**0.00**	**[-0.92, 0.02]**
			800-2000	**0.02**	**0.04**	**0.00**	**[-0.01, 0.02]**
	Nordic Seas		0-5	0.07	0.15	-0.00	[-0.09, 0.08]
			5-100	0.05	0.09	-0.01	[-0.06, 0.03]
			100-300	0.03	0.05	-0.00	[-0.03, 0.03]
			300-800	0.02	0.03	0.01	[-0.00, 0.02]
			800-2000	0.01	0.01	0.00	[-0.00, 0.02]

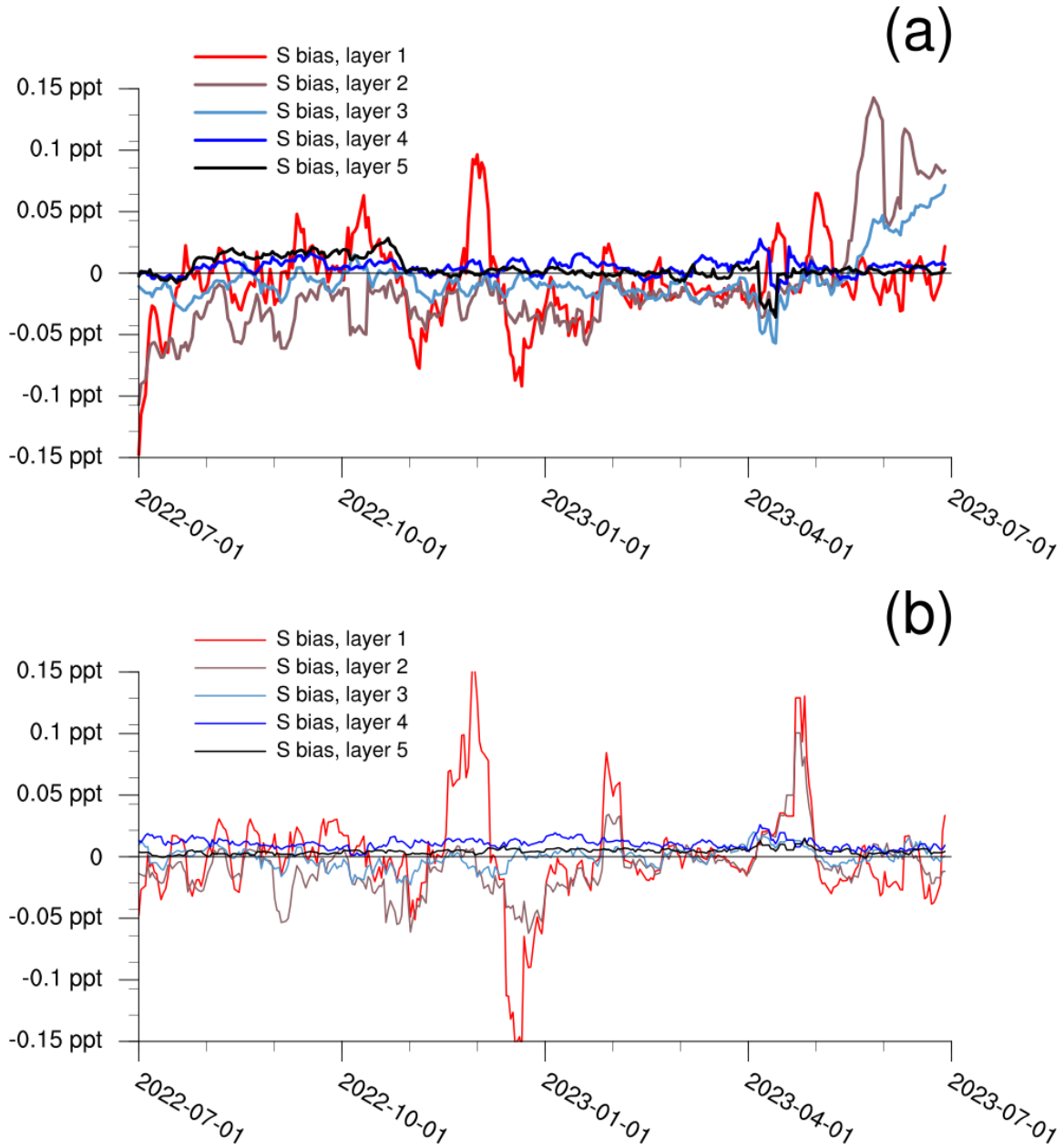


Figure 17: *Temporal evolution of the bias in the model results for salinity, for the five layers for (a) the full domain and (b) the Nordic Seas sub-domain. Daily results, after smoothing with a seven-day boxcar filter.*

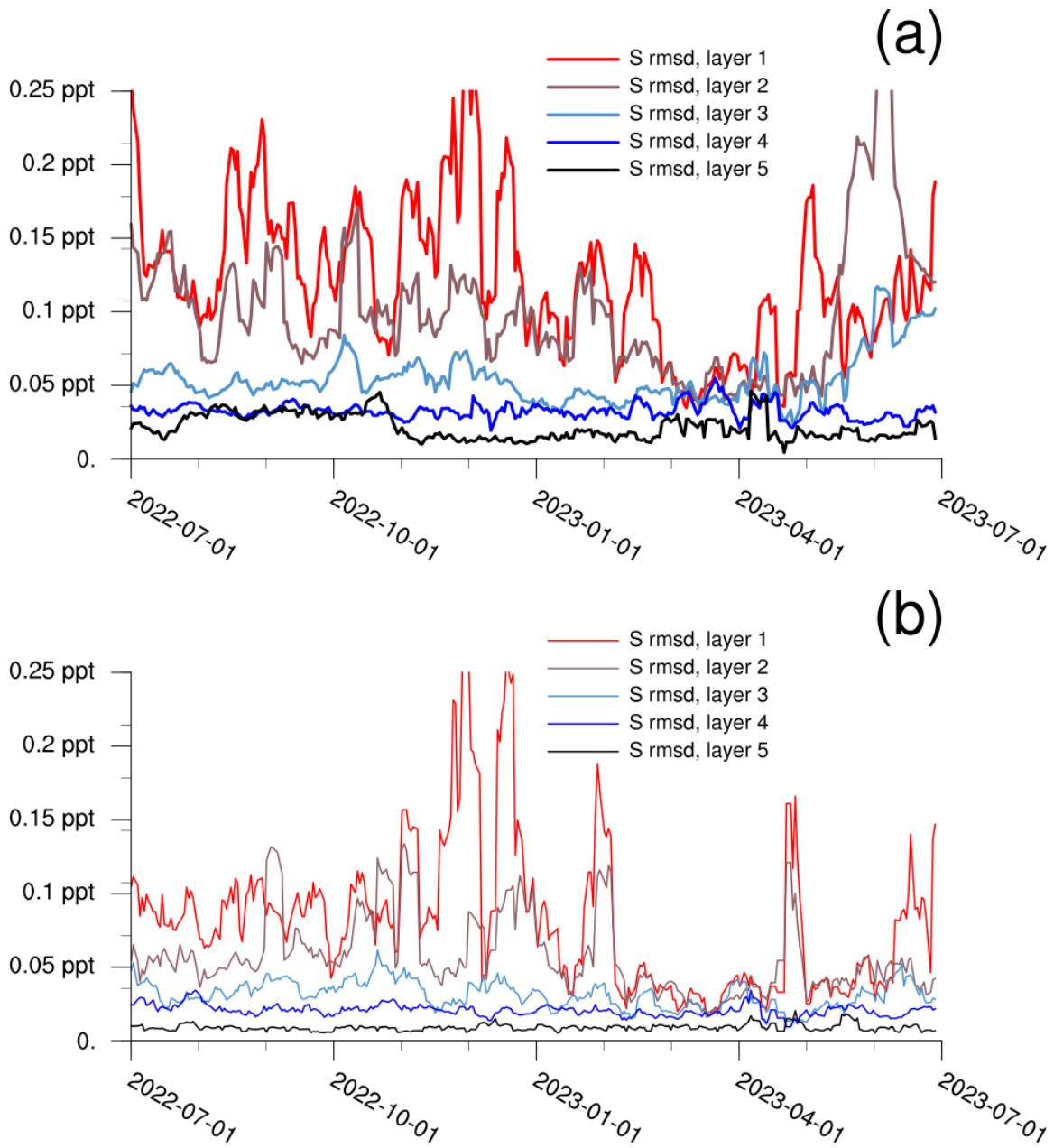


Figure 18: *Temporal evolution of the RMSD of the model results for salinity, for the five layers for (a) the full domain and (b) the Nordic Seas sub-domain. Displayed here are daily results, after smoothing with a seven-day boxcar filter.*

***Table 20**: Estimated accuracy numbers for the mixed layer depth. Units: m. Results are based on observations taken at positions shown in Figure 14. See Section III.3 for an explanation of the percentiles. The Nordic Seas domain is shown in Figure 1(b).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Mixed layer depth (m)	**Full domain**	**Profiling instruments**	**52**	**156**	**20**	**[-21, 127]**
	Nordic Seas		36	99	3	[-39, 71]

IV.8 Mixed layer depth (MLD)

MLD is assessed using the same dataset as for salinity and potential temperature (Sections IV.6.2 and IV.7). The profile positions are displayed in Figure 14.

We applied a density-dependent definition of MLD. Hence, MLD is not directly observed but derived from the profile data for salinity and potential temperature. We use the Gibbs Sea Water Functions (Pawlowicz et al., 2012) for this purpose. We first discard all observations from above 10 m to remove any impact of near-surface processes. We then define MLD as the vertical level were the increase in density exceeds 0.03 kgm^{-3} from its value at the uppermost level (below 10 m).

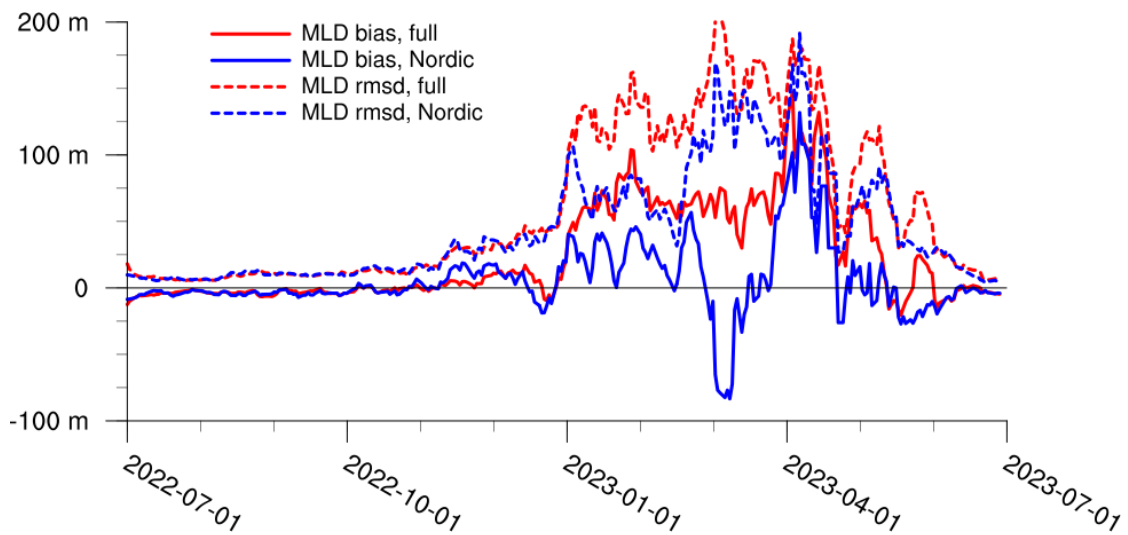


***Figure 19:** Temporal evolution for the bias and RMSD of mixed layer depth (MLD). Daily results, after smoothing with a seven-day boxcar filter.*

The results in Table 20 reveal that the quality of the model results in the Nordic Seas are notably better than the full domain results, as e.g. the bias is 3m for the former region, and 20m for the latter. We also note from the percentile results that the variability is considerable, by a factor of 3 for the RMSD differences.

The large variability is easily confirmed by the metrics' timeseries in Figure 19. We note that the seasonal signal is exceptionally strong, with model results and observations being nearly identical during summer and autumn, but exhibiting large deviations during winter and spring, with RMSD values of the order of 100m.

This finding is closely related to the seasonal signal in the mixed layer depth, since it can span more than 1000m in the Nordic Seas (blue region in Figure 1b, from 61°N to 79°N) due to weak stratification during winter and spring. Then, small to moderate differences between the profiles from the model and observations can lead to substantial values for the mixed layer depth quality metrics. As an example, a temperature difference of 0.1K (weak stratification) corresponds to a density difference of 0.02 kg/m^3, i.e. inside the density span which defines the mixed layer depth.

The tendency for a larger deep mixed layer in the model results compared to the field observations in the full domain is clear from the full red line in Figure 19, which is rarely negative. Unfiltered time series exhibit even stronger fluctuations than the lines in this figure, given the application of a seven-day boxcar filter.

IV.9 Sea level anomaly

Sea level anomaly is assessed based on observations from satellite altimetry (Sentinel-6a, Jason-3 and Cryosat-2).

Prior to comparing model results and observations, the model sea level is modified by a time invariant field that describes the differences in sea surface height between the model results and the observed mean for the 20-year period from 1993 to 2012.

***Table 21**: Estimated accuracy numbers (in cm) for sea level anomaly See Section III.3 for an explanation of the percentiles. The Nordic Seas domain is shown in Figure 1(b).*

Period: July 2022 – June 2023						
Variable	**Region**	**Supporting observations**	**RMS difference**		**Bias (model-observation)**	
			Mean	90%-ile	**Mean**	5-95%-ile range
Sea level anomaly (cm)	**Full domain**	**Sentinel-6a**	**7**	**8**	**-2**	**[-4, 1]**
		Jason-3	**7**	**8**	**-2**	**[-4, 1]**
		Cryosat-2	**8**	**10**	**-2**	**[-5, 2]**
	Nordic Seas	Sentinel-6a	4	6	1	[-4, 5]
		Jason-3	4	6	0	[-4, 5]
		Cryosat-2	5	7	1	[-3, 6]

The temporal variability of the RMSDs is displayed in Figure 20. The results reveal that the quality metrics have low seasonal variability, usually inside 10%-20% of the magnitude of the annual mean value. Furthermore, this depiction confirms the results given in Table 21, where the mean of the daily RMSD values for the Nordic Seas sub-domain was found to be lower than the corresponding result for the full domain. Both groups of results only exhibit a weak seasonality.

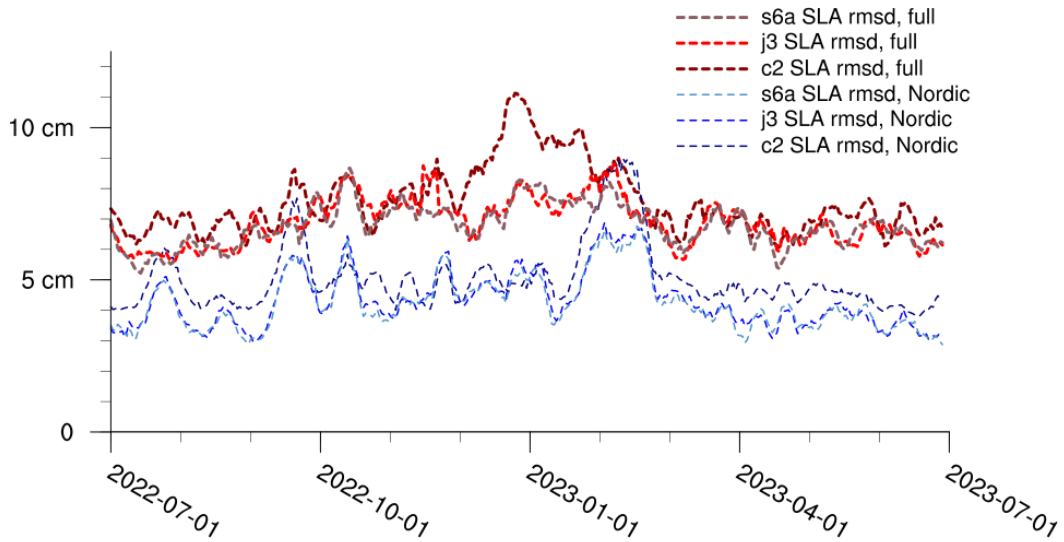


***Figure 20**: Temporal evolution of the RMSDs between model results and observations of sea level anomalies. Daily results, after application of a seven-day boxcar filter.*

IV.10 Vertical velocity: visual inspection

The results for the vertical velocities cannot be validated against near-real-time measurements thus this validation is only a qualitative comparison against the global high-resolution product on a randomly chosen day.

From Figure 21, we note that the products show some similarities: the absence of vertical velocities below the Arctic sea ice cover, the upwelling feature in the East Greenland Sea related to a passing storm and the reasonable magnitude of the vertical velocities of a few meters per day.

There are also qualitative differences in the density of small-scale features of vertical velocities, which are about 3 times denser in the global than the Arctic product. In the absence of validation data, it cannot be concluded which of the two products is more correct. Several aspects can explain the differences, like the different vertical coordinate system of the two products (z-isopycnic versus isopotential) and therefore the vertical velocities due to associated differences in the numerical discretisations of the governing equations, and the lower horizontal resolution of the Arctic product (6 km against 4 km). Comparisons in deeper waters (1000 m) for the sample date displayed in Figure 21 reveal lower average vertical velocities in the Arctic product (1.8 m/day) than in the global product (4.2 m/day).

Overall, the vertical velocities are deemed realistic by comparison to another model.

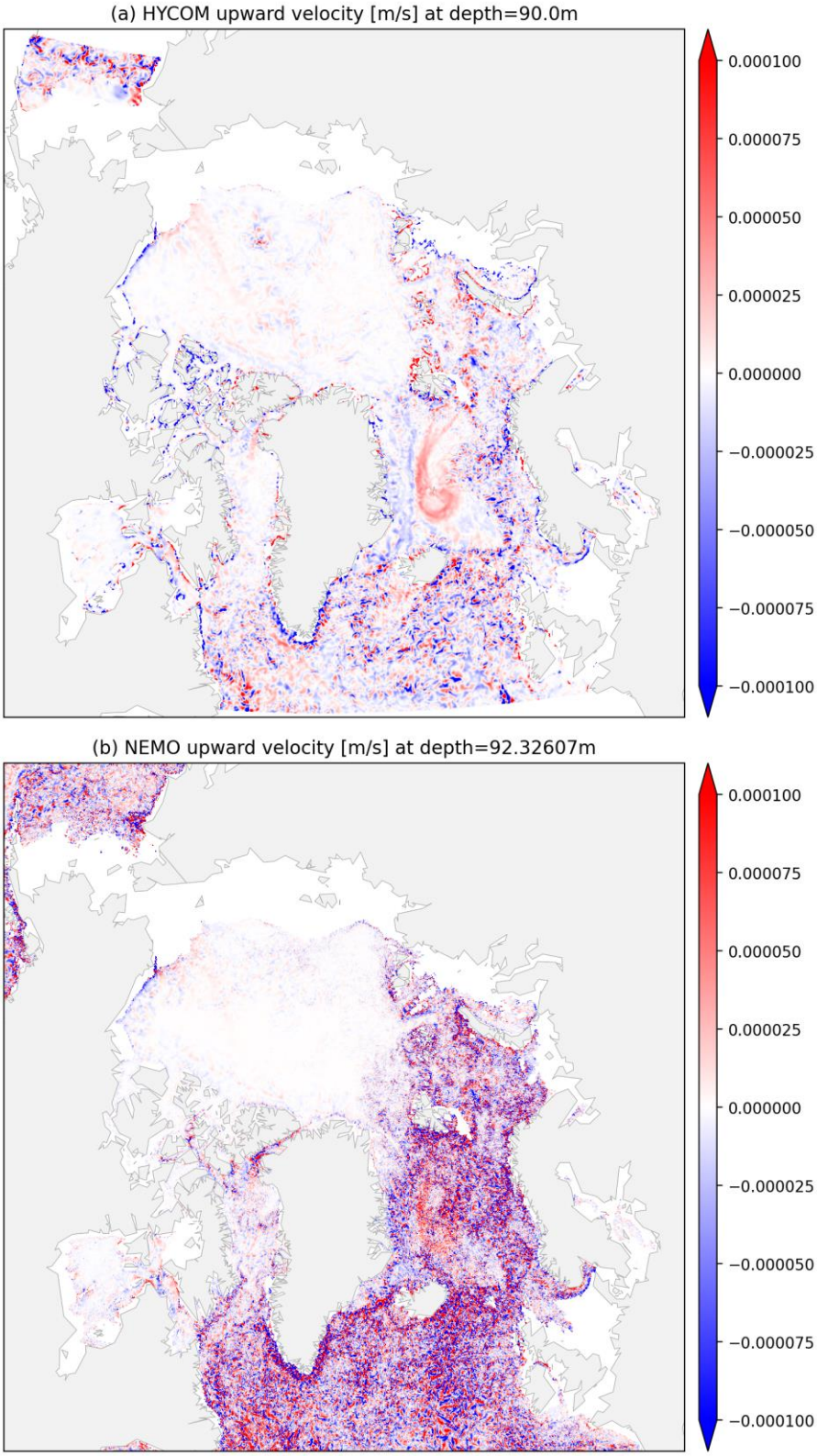


Figure 21: *Daily averaged vertical velocities at 90 m depths on a representative day (15th May 2023). (a): from the ARC MFC forecast product. (b): from the corresponding GLO HR MFC product.*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This document describes the first version of the Arctic MFC 6.25 km ocean and sea ice forecasting system. The present section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

***Table 22**: List of events, outages and/or changes for each version of the present product.*

Date	Change/Event description	System version	Other
15/9/2023	First version of the TOPAZ5 forecast version, using HYCOM v.2.2.98, CICE v.5.1, NERSC DEnKF v.2.11 for data assimilation, and the ESMF v.8.0.1 coupler.	1.0	
2/10/2025	River runoff implemented as real time daily results from the Copernicus GloFAS product. This replaces monthly runoff climatology from HYPE.	1.1	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between previous and current version.

May 2025:

The only change from the previous version is in the implementation of river runoff. We have reason to believe that the new implementation will lead to improved results, mainly due to a better representation of runoff temporal variability. However, observations of salinity close enough to river mouths to quantify this improvement is not available.

September 2023:

This is the first version of ARCTIC_ANALYSISFORECAST_PHY_002_001. This product replaces ARCTIC_ANALYSIS_FORECAST_PHYS_002_001_a. The quality changes in the new product are:

- Results for the sea ice concentration have improved somewhat
- Biases in the position and length of the sea ice edge have improved
- Sea ice drift results reveals an improvement for the length of the 1-day drift
- SST results have improved slightly
- Temperature profiles have improved in the entire water column
- Salinity profiles have improved in the entire water column
- Mixed layer depth results have improved in the Nordic Seas

These findings are based on comparisons of results from a best estimates ensemble with the new configuration, and the results for the first day of forecasts with the previous configuration (forecasts from the new configuration are not yet available).

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